

Labor Supply Shocks and Child Labor among U.S. Farm Households: Evidence from the Bracero Termination

Abstract

The 1964 termination of the Bracero program abruptly ended Mexican seasonal farm labor in the U.S., representing one of the largest permanent contractions in immigrant farm labor in U.S. history. Existing work finds little evidence that domestic workers filled the gap. I focus on a previously unexplored margin: child labor within farm households. Under the Fair Labor Standards Act, children on family farms face no minimum age requirement and no hours restrictions, an exemption unique to agriculture. Using decennial Census microdata and a continuous difference-in-differences design exploiting cross-state variation in Bracero reliance, I find that the termination shock significantly increased child farm labor participation and hours worked. School attendance declined among boys, but educational attainment showed no measurable decline.

1 Introduction

Whether immigration restriction can improve labor market conditions for domestic workers is among the most contested questions in economics, and among the least well answered. The canonical test comes from the termination of the Bracero program in 1964, one of the largest deliberate contractions of immigrant labor in U.S. history. At its peak, the program had supplied nearly half a million Mexican seasonal workers to U.S. farms each year, and its termination was executed with the explicit goal of raising wages and employment for domestic farm workers. Clemens et al. (2018), in the most comprehensive analysis of this episode, find no detectable effect on either, with adjustment occurring instead through mechanization and shifts in crop mix. This implies that domestic labor supply in agriculture is largely inelastic to immigration restrictions.

Yet Clemens et al. (2018) and most of the agricultural labor market literature focus almost exclusively on adult workers. A different and largely unstudied margin of domestic labor supply adjustment is child labor within farm households. Under the Fair Labor Standards Act (FLSA), children working on family farms face no minimum age requirement and no restriction on hours worked, even during school hours, an exemption far broader than those available in any other sector of the U.S. economy. This legal environment means that farm families can freely deploy their children in response to shocks that reduce the availability of adult hired workers, and that the null adult substitution results in the existing literature may partly reflect labor adjustment occurring through a margin not captured in standard employment and wage data.

This paper asks whether the termination of the Bracero program increased child labor supply on U.S. family farms and whether any such increase came at a cost to children's educational attainment. I use decennial Census microdata for children aged 14-17 in farm households across five Census years spanning 1940 to 1980. I construct a continuous treatment variable defined as the interaction of each state's pre-termination Bracero worker share with an indicator for post-1964 Census years, which captures the intensity of the labor

supply shock facing farm households in each state. I then estimate a continuous difference-in-differences specification exploiting variation in pre-termination Bracero reliance across states. Identification relies on the assumption that, absent the termination, child labor and education outcomes would have trended similarly across states regardless of their pre-termination Bracero reliance.

I find robust evidence that Bracero termination increased child farm labor participation. Estimates indicate that moving from zero to full pre-termination Bracero reliance increased the probability that children aged 14-17 in farm households were employed as farm laborers by approximately 15 percentage points and the probability of reporting any positive work hours by approximately 19 percentage points after 1964. Hours worked also increased along the intensive margin. School attendance declines significantly following termination, while grade-for-age and highest grade attained remain statistically indistinguishable from zero, suggesting that Bracero termination induced measurable schooling disruption that did not translate into lasting losses in educational attainment. These effects are supported by event study evidence showing no differential pre-trends prior to termination, with the labor supply response concentrated in the period immediately following termination and attenuating in the subsequent decade as farms mechanized and alternative labor sources emerged. A falsification test shows no effect among children in non-farm households in the same states, confirming that the results reflect the farm labor supply channel rather than broader state-level economic trends. I also rule out selection into farming in response to termination, which could alter the sample composition and drive the estimates.

Heterogeneity analysis reveals that both the labor supply and schooling responses are concentrated among boys and older teenagers. Estimates indicate that moving from zero to full pre-termination Bracero reliance increased the probability that boys aged 14-17 were classified as farm laborers by approximately 25 percentage points following termination and the probability of reporting any positive work hours by 29 percentage points, while girls show smaller and less precisely estimated responses on formal labor supply measures.

Girls do show modest increases in hours worked and any-work-hours participation, though without a corresponding increase in the farm laborer indicator, suggesting they were deployed in informal and intermittent tasks that do not register in formal occupation classifications. School attendance declines significantly among boys but not girls, indicating that the human capital cost of Bracero termination was borne asymmetrically along gender lines, a pattern consistent with the informal and intermittent nature of girls' labor involvement not accumulating into measurable schooling disruption. Labor supply effects also increase monotonically with age from 10 percentage points at age 14 to 18 percentage points at age 16, and the school attendance decline is concentrated among 17-year-olds. The results are robust to alternative treatment definitions, specifications, and sample restrictions.

This paper contributes to several strands of literature. First, it contributes to the literature on the effects of labor supply shocks on U.S. agricultural labor markets. A growing body of work uses the Bracero termination and more recent enforcement shocks to study how farm labor markets adjust when immigrant labor supply contracts (Charlton and Kostandini, 2021; Clemens et al., 2018; Ifft and Jodlowski, 2022; Kostandini et al., 2014; Luo and Kostandini, 2022; Richards, 2018; Rutledge and Mérel, 2023; San, 2023; Zahniser et al., 2012). This literature focuses primarily on wages, employment levels, mechanization, and crop choice as margins of adjustment, and consistently finds limited substitution by domestic adult workers. This paper extends this literature by showing that farm households increased their reliance on child labor after Bracero termination, identifying an additional channel through which agricultural labor markets adjusted to immigrant labor scarcity.

Second, this paper adds to the literature on the household welfare effects of immigration policy. A growing literature documents how immigration enforcement and restrictions affect the welfare of immigrant households, including effects on children's health and educational outcomes (Amuedo-Dorantes and Antman, 2017; Amuedo-Dorantes et al., 2018; Hainmueller et al., 2017; Martinez-Donate et al., 2020; Monras, 2020). A related strand shows that enforcement also reshapes labor allocation within native households, as

reductions in immigrant labor supply in the childcare sector constrain native mothers' labor market participation (Ali et al., 2024; East, 2020). Yet the consequences of immigration restrictions for child labor within native farm households, and the implications for children's educational attainment, remain unexplored. This paper suggests that immigration-induced labor scarcity altered intra-household labor allocation within native farm families, increasing children's labor supply and disrupting school attendance.

Third, this paper contributes to the limited literature on child labor and schooling in the United States. While child labor and its educational consequences are extensively studied in developing country contexts, where agricultural shocks are shown to increase child labor (Beegle et al., 2006; Colmer, 2021; Dammert, 2008; Kruger, 2007), the U.S. literature remains thin, particularly for the post-FLSA era. Existing U.S. work focuses primarily on the early twentieth century (Manacorda, 2006; Moehling, 1999). The only work on child labor in U.S. agriculture is descriptive: Fan et al. (2014) document that child agricultural labor remains substantial in the modern United States and that the FLSA exemption has left young farm workers vulnerable. A related body of work examines how agricultural shocks affect schooling and occupational outcomes in rural America. For instance, Baker (2015) and Baker et al. (2020) document that the boll weevil infestation reduced educational attainment in the rural South, while Liu (2026) shows that the Dust Bowl reshaped intergenerational occupational choices within farm households. These papers establish that agricultural disruptions have meaningful and lasting effects on farm household decisions, but do not study child labor supply directly. This paper bridges these strands by linking a major agricultural labor market shock to child labor and schooling outcomes within U.S. farm households, showing how labor scarcity can shape short-run human capital investment decisions.

The remainder of the paper proceeds as follows. Section 2 provides background on the Bracero program and child labor regulation. Section 3 describes the data, and Section 4 outlines the empirical framework. Section 5 presents the results, and Section 6 concludes.

2 Background

2.1 The Bracero Program

The Bracero program, formally the Mexican Farm Labor Agreement, was established in 1942 as a wartime measure to address farm labor shortages following the entry of the United States into World War II. Under the program, the U.S. and Mexican governments jointly administered the recruitment and placement of Mexican agricultural workers, primarily for seasonal crop work. Workers entered under bilateral contracts that specified wages, working conditions, and return obligations, and were deployed primarily in labor-intensive crop production including fruits, vegetables, cotton, and sugar beets.

From its wartime origins, the Bracero program grew substantially in scale through the 1950s, reaching a peak of approximately 445,000 workers in 1956 (Calavita, 2010). By this point, Bracero workers accounted for a substantial share of the seasonal hired agricultural workforce in many states, and their presence had become deeply embedded in the labor practices, cropping patterns, and farm organization of the regions most reliant on them. The program was repeatedly renewed by Congress despite persistent criticism from domestic labor advocates, religious organizations, and civil rights groups who documented wage suppression, poor working conditions, and displacement of domestic farm workers.

The program's termination in December 1964 was abrupt and politically driven. Following a CBS documentary in 1960 that documented the conditions of domestic migrant workers, domestic labor unions led by the American Federation of Labor and Congress of Industrial Organizations and the emerging farmworker movement under Cesar Chavez intensified pressure on Congress to end the program. Public Law 78, the legislative basis for the Bracero program, was allowed to expire on December 31, 1964 without renewal. The termination was nationwide and permanent, representing a sharp and durable contraction in the legal supply of Mexican farm workers precisely in the regions most dependent on them.

Table A.1 documents the crop-level concentration of foreign seasonal workers in 1964, the year of program termination. Foreign workers accounted for more than half of seasonal workers in lettuce (55.3%) and nearly half in sugarcane (46.9%), with substantial shares also in celery (32.4%), cucumbers (27.4%), melons (28.4%), and tomatoes (26.2%). By contrast, foreign worker shares were negligible in tobacco (1.9%), beans (2.4%), and cotton (3.7%). This crop-level variation suggests that Bracero termination created a highly uneven labor supply shock concentrated in labor-intensive fruit and vegetable crops that were difficult to mechanize.

Figure 1 documents the rise and fall of the Bracero program over time. Panel (a) plots the monthly stock of Mexican foreign farm workers nationally. The number of Mexican Bracero workers grew from near zero in 1942 to a peak of approximately 300,000 in the late 1950s before declining sharply after 1962 as political pressure to end the program intensified, reaching near zero by 1965.¹ The strong seasonality in Mexican farm worker presence reflects the harvest-intensive nature of the crops they worked, primarily fruits, vegetables, and cotton, which concentrated labor demand in summer and fall months.² Non-Mexican foreign workers, primarily from Jamaica, the British West Indies, and Canada, remained a small and relatively stable presence throughout the period and did not substitute for the sharp decline in Mexican workers following termination (see Figure A.1).

Panel (b) plots the average Mexican share of hired seasonal farm workers by year, separately for high and low Bracero reliance states. The average Mexican share in high-reliance states reached approximately 30% at peak utilization in the late 1950s, compared to under 3% in low-reliance states, underscoring the sharp geographic concentration of treatment intensity that drives identification in this paper.

The exposure measure varies substantially across states. Figure A.2 maps the geo-

¹The temporary contraction between 1947 and 1951 reflects the end of World War II and the return of American men to the domestic labor force, which reduced demand for foreign farm workers.

²Because workers entered and exited throughout the agricultural season, the total number of Bracero workers employed over an entire year was substantially larger than the peak monthly stock shown in the figure.

geographic distribution of Bracero worker counts across the 46 states in the analysis sample. Texas and California had the highest pre-termination Bracero reliance, with an average of approximately 59,150 and 48,325 Mexican Bracero workers respectively in 1955. New Mexico and Arizona also had substantial Bracero presence, while states in the Midwest and Southeast had little or no exposure. This geographic variation is driven primarily by proximity to Mexico, the prevalence of labor-intensive crops such as fruits and vegetables, and the historical recruitment networks that the program established in its early years.

Figure 2 plots the Bracero share of total hired seasonal farm workers in 1955, the peak Bracero year. The geographic pattern shifts somewhat from the count map: New Mexico had the highest pre-termination Bracero share at 60.9% of hired seasonal farm workers, followed by Nebraska (32.3%), Arizona (26.9%), Texas (24.1%), and California (23.1%). This shows that some smaller agricultural states were heavily dependent on Bracero labor relative to their total farm workforce even if their absolute numbers were modest. Twenty-three states had no Bracero workers in 1955. Rhode Island and New Hampshire, shaded in grey, are excluded from the analysis due to missing farm worker stock data in the original sources. Table A.2 reports state-level Bracero share measures for all 46 states in the analysis sample.

Clemens et al. (2018) provide the most comprehensive empirical analysis of the Bracero termination's effects on U.S. farm labor markets. Using administrative program records matched to Farm Labor Survey data, they show that termination reduced farm employment in heavily exposed states and accelerated mechanization, while finding limited evidence of substantial domestic labor market adjustment through wages or adult worker substitution. These findings motivate the possibility that adjustment may instead have occurred through within-household labor substitution margins. Related work by San (2023) documents that labor scarcity induced by the termination accelerated labor-saving mechanization in cotton production, consistent with induced innovation mechanisms.

2.2 Child Labor Regulation and the FLSA Agricultural Exemption

The Fair Labor Standards Act of 1938 established federal minimum wage, overtime, and child labor standards for U.S. workers. However, the FLSA included a specific and broad exemption for agricultural employment. Under the agricultural exemption, children of any age may work on a farm owned or operated by their parents, with no restriction on hours or timing relative to school schedules. Children as young as 12 may work on any farm with parental consent, and children aged 14 and older may work in any non-hazardous agricultural occupation.

In the institutional context of the 1960s and 1970s, these exemptions were even broader than they are today. Enforcement of compulsory attendance laws was weak in rural areas, and the social and economic organization of farm households assumed substantial contributions of family labor including children. This institutional environment means that farm families faced no legal barrier to substituting child labor for adult hired labor when access to Bracero workers was withdrawn. The persistence of child farm labor under this legal framework is documented in the modern context by Fan et al. (2014), who find that child agricultural labor remains substantial in the United States despite some decreases over time, and argue that the FLSA agricultural exemption has left young workers vulnerable to ongoing labor demands. The degree to which they did so is the central empirical question of this paper.

3 Data

3.1 Census Microdata

I use decennial Census microdata from IPUMS-USA (Ruggles et al., 2024) for five Census years: 1940, 1950, 1960, 1970, and 1980. The decennial Census is particularly well-suited to this question for three reasons. First, it provides large enough samples of children in

farm households to estimate effects with reasonable precision, a population that is too thin in most survey data to support the kind of subgroup analysis required here. Second, it covers the full pre- and post-termination period at regular intervals, allowing me to assess both pre-existing trends and the persistence of the treatment effect across multiple post-termination decades. Third, it is the only source providing population-level coverage of child labor and schooling outcomes for this period.

The 10-year interval means the estimates capture medium- to long-run equilibrium responses rather than immediate adjustment. This approach follows Boustan (2010) and Stephens Jr and Yang (2014) in using repeated Census cross-sections to study responses to major policy shocks, though the absence of annual data between 1964 and 1970 means the estimates cannot speak to how the response evolved in the years immediately following termination.

I restrict the sample to children aged 14-17 residing in farm households, which the Census identifies as households living on a farm. The lower bound of 14 reflects both a substantive and data quality consideration: in the Census, the hours worked and occupation questions were administered only to individuals in the labor force, meaning children under 14 are assigned zero by construction rather than from genuine survey responses.³ The upper bound of 17 also ensures that I am studying children who are still of school age and for whom the school-work tradeoff is relevant. I further restrict the sample to U.S.-born children to focus on the domestic population whose labor supply is the object of study, excluding the children of immigrant farm workers whose labor supply decisions may respond differently to the Bracero termination.

The primary labor supply outcomes are a farm laborer indicator, equal to one if the child's occupation is coded as farm laborer, wage farm worker, or unpaid family farm worker; an any-work-hours binary, equal to one if the child reports positive hours worked

³This pattern is consistent with the FLSA agricultural exemption, which permits farm work beginning at age 14 with parental consent on any farm, suggesting that the Census universe restriction and the institutional environment jointly produce the sharp onset of farm labor participation at age 14.

in the reference week; and hours worked last week as a continuous measure.⁴ The farm laborer indicator identifies children whose primary activity is farm work, while the any-work-hours binary captures broader labor force participation, including children who work part-time alongside school attendance. The continuous hours measure captures the intensive margin of labor supply.

The primary schooling outcomes are school attendance, grade-for-age, an indicator equal to one if the child's highest grade attended is at least the expected grade for their age, constructed as $higrade \geq age - 6$, and highest grade attended. I examine three schooling outcomes that capture different dimensions of educational investment. School attendance reflects current participation but is a weak proxy for actual educational engagement in this context.⁵ The Census records only whether a child is attending school, not whether they attend regularly, how many days they are absent, or how they perform academically. These unobserved margins, attendance, absenteeism, and academic achievement, are arguably more sensitive to seasonal farm work demands than formal attendance, since a child can remain attending while missing school during harvest periods. This is an important limitation of the Census data and implies that my null schooling results should be interpreted as the absence of effects on formal attendance and grade progression, not as evidence that farm work had no effect on actual learning or school attendance. Grade-for-age captures cumulative educational progression and is more sensitive to disruptions that

⁴The Census records hours worked in brackets rather than exact values. I assign each observation the midpoint of its reported bracket to construct the continuous hours variable.

⁵The decennial Census interviews were conducted primarily in April. The definition of schooling shifts modestly across years: 1940 and 1950 count vocational and night school if part of the regular school system, while 1960 explicitly excludes trade and business schooling. For 14-17 year olds in farm households, the 1960 change is the most relevant, though its practical effect on measured attendance at this age is likely small. The school attendance question in each Census year refers to attendance since February 1 (March 1 in 1940). Because this reference period is consistent across all Census years, any differential attendance decline detected across high and low Bracero states is not an artifact of differential measurement timing, and the significant decline documented here is best interpreted as a lower bound on total schooling disruption given that attendance during peak harvest months is not captured. Similarly, because the interview reference week falls in spring rather than during peak summer and fall harvest months, the hours worked measure likely understates total annual child farm labor, reinforcing the interpretation of all labor supply estimates as lower bounds on the true response.

cause children to fall behind even if they remain attending.⁶

Table 1 presents summary statistics for the main analysis sample of U.S.-born children aged 14-17 in farm households. Farm households account for approximately 12.5% of the full sample of U.S.-born children aged 14-17 across the five Census years. Farm labor participation is substantial: approximately 19% of farm children aged 14-17 are classified as farm laborers and approximately 20% report any positive work hours in the reference week, with average hours worked of 0.784 per week including the large share of non-working children. Among children who report positive hours, Figure A.3 shows that among working farm children, hours are distributed across the full range of categories, with notable concentrations at part-time hours (1-29 hours) as well as at full-time and above (40 hours or more), suggesting heterogeneity in how farm families deployed child labor in response to the termination shock. School attendance is high at 80% and approximately 87% of farm children are on track in grade progression for their age. Highest grade attended averages 11.1 years across the 14-17 age range.

Farm households are large, averaging six persons per household. The racial composition of the farm child sample is notable: Black children account for 13.7% of the sample, substantially above their share in the general population, reflecting the historical concentration of Black farm families in the South. Latino children account for only 1.4% of the sample, which likely reflects the restriction to U.S.-born children, excluding the children of immigrant farm workers who would disproportionately be Latino. The sample is approximately evenly split by gender at 47.6% female.

⁶The 1970 Census microdata present a data availability challenge because the school attendance question was administered on Form 2 of the Census questionnaire rather than Form 1. For labor supply outcomes, I pool the 1970 Form 1 and Form 2 state samples and scale person weights by 0.5 to reflect the combined 2% sampling rate. For schooling outcomes, I restrict the 1970 sample to Form 2 only, as school attendance was not asked on Form 1. For all other Census years, I use the largest available samples. Sample weights are applied throughout.

3.2 Bracero Exposure Measure

I measure pre-termination Bracero reliance using the state-level Bracero worker share constructed by Clemens et al. (2018) from administrative records of the U.S. Department of Labor and the Bureau of Employment Security. Clemens et al. (2018) assembled these data through in-person visits to government archives in Washington, D.C. and presidential archives in Abilene, Kansas and Independence, Missouri, collecting primary print sources on monthly stocks of hired farm workers by state that had not previously been compiled into a secondary dataset. The records provide monthly counts of Bracero workers placed in each state under the program from 1943 to 1973, matched to data on total hired seasonal farm employment from the Farm Labor Survey conducted by the U.S. Department of Agriculture.

The underlying farm survey used the ES-223 form administered by state-level employment service offices, which explicitly defined foreign workers as those who “legally entered the continental United States but who normally reside in a foreign country,” with instructions that “illegal entrants are not to be included.” The Bracero share measure therefore captures only the legal immigrant labor supply shock, isolating the policy-induced contraction in authorized seasonal labor from any concurrent changes in unauthorized employment. The validity of the measure is further supported by the close correspondence between national totals reported in the farm surveys and independent records of bracero departures compiled by the Mexican government, suggesting the survey adequately covered establishments employing Mexican seasonal labor (Clemens et al., 2018, Online Appendix A5.7).⁷

Following Clemens et al. (2018), the treatment intensity variable is the fraction of total hired seasonal farm workers who were Mexican Bracero workers in 1955, constructed as the average monthly share across all months of 1955, and denoted $\bar{m}_s$ for state s . I use

⁷Farm worker stock data are missing in the original sources for Rhode Island and New Hampshire in 1955. Alaska and Hawaii are excluded as non-contiguous states with no Bracero program presence. These four states are therefore excluded from the analysis sample, leaving 46 states.

the 1955 share for two reasons. First, it falls near the peak of Bracero utilization, when the program was fully operational and deeply embedded in agricultural labor markets, before the political environment regarding immigrant farm workers began to shift in the early 1960s. Second, it ensures the measure is predetermined relative to both the 1964 termination and any anticipatory behavioral responses by farm households, avoiding the mechanical endogeneity that would arise from measuring exposure closer to 1964. I verify that results are robust to using the average Bracero share across 1954-1963 as an alternative exposure measure, which further guards against any influence of anticipatory adjustments in the years immediately preceding termination.

4 Empirical Framework

4.1 Difference-in-Differences Specification

My primary estimating equation is a continuous difference-in-differences specification:

$$y_{ist} = \beta (\overline{m}_s \times \mathbf{1}[t \geq 1965]) + \mathbf{X}'_{ist}\gamma + \delta_s + \tau_t + \varepsilon_{ist} \quad (1)$$

where y_{ist} is the outcome for child i in state s in Census year t ; $\overline{m}_s$ is the pre-termination Bracero share for state s ; $\mathbf{1}[t \geq 1965]$ is an indicator for Census years after the program's termination (1970 and 1980 in my sample); $\mathbf{X}_{ist}$ is a vector of individual and household controls including household size, age, gender, and race/ethnicity indicators, and birth rank among children under 18 in the household (where rank 1 denotes the eldest), as well as household head characteristics including the head's age, sex, marital status, and highest grade attended; δ_s are state fixed effects that absorb time-invariant differences across states in child labor practices, farm structure, and labor market conditions; and τ_t are year fixed effects that absorb national trends in child labor, schooling, and agricultural employment. Standard errors are clustered at the state level to account for the fact that the treatment

variable $\overline{m}_s$ varies at the state level. All specifications are estimated using Census person weights.⁸

Since $\overline{m}_s$ is a fraction bounded between zero and one, β represents the effect of moving from a state with no Bracero presence to a state where Bracero workers constituted the entire hired seasonal farm workforce. A positive β on labor supply outcomes indicates that Bracero termination increased child farm labor participation, while a negative β on schooling outcomes would indicate a reduction in educational investment, both consistent with farm families substituting toward child labor when access to immigrant workers was withdrawn.

The treatment variable is measured at the state level, reflecting the geographic unit at which Bracero worker placements were recorded in the administrative sources compiled by Clemens et al. (2018) and the level at which deployment was concentrated within particular agricultural regions. Finer geographic variation is not available in the original sources. Estimates should therefore be interpreted as average effects across farm households, scaled by state-level Bracero exposure, which likely attenuates heterogeneity within states.

4.2 Identification

Identification of β relies on the assumption that, absent the Bracero termination, trends in child farm labor would have been parallel across states with different doses of pre-termination Bracero reliance. The state fixed effects absorb any time-invariant differences across states, while the year fixed effects absorb common time trends. The remaining variation exploited for identification is the differential evolution of outcomes across states after 1964, proportional to their pre-termination Bracero exposure. I assess the validity of this assumption through event study specifications, which allow the treatment effect to vary freely across Census years and test for pre-existing differential trends prior to 1964. The

⁸For labor supply outcomes, person weights are scaled by 0.5 for 1970 observations to reflect the pooling of Form 1 and Form 2 state samples. For schooling outcomes, the 1970 sample is restricted to Form 2 only and original Census person weights are used.

reference year is 1960, the last Census year before the termination. Pre-trend coefficients for 1940 and 1950 that are statistically indistinguishable from zero provide support for the parallel trends assumption. Specifically, I estimate:

$$y_{ist} = \sum_{t \neq 1960} \alpha_t (\bar{m}_s \times \mathbf{1}[\text{year} = t]) + \mathbf{X}'_{ist} \gamma + \delta_s + \tau_t + \varepsilon_{ist} \quad (2)$$

where the coefficients α_t trace the differential evolution of outcomes across states, proportional to their pre-termination Bracero exposure, relative to 1960. A positive and significant α_{1970} indicates that Bracero termination increased child farm labor in the first post-termination Census year.

In the language of shift-share designs, the nationwide termination of the Bracero program constitutes the common shock, while the 1955 Bracero share captures differential treatment intensity across states, so that identification comes from cross-sectional differences in pre-existing reliance rather than from endogenous post-treatment adjustments. The shift-share structure of the treatment variable requires that the 1955 Bracero share is uncorrelated with state-level trends in child labor that would have emerged after 1964 absent termination (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020). This assumption is supported by the fact that Bracero exposure was driven primarily by proximity to the Mexican border and historical recruitment networks established in the program's early years, factors plausibly orthogonal to child farm labor trajectories.

To the extent that unauthorized Mexican migration partially offset the labor supply shock after termination, my estimates capture the net effect of the policy change inclusive of this margin of adjustment and represent a lower bound on the effect of immigrant labor scarcity in the absence of any such response. However, the evidence against meaningful unauthorized substitution in the immediate post-termination period is strong. Based on Mexican government departure and return records, Bracero overstay rates were under 1% in both 1963 and 1964, and Border Patrol apprehensions of Mexican nationals barely rose in the

years immediately following termination, with large undocumented inflows concentrated in the 1970s and 1980s rather than the 1960-1970 window (Clemens et al., 2018). Total Mexican farm worker counts also fell meaningfully following termination, consistent with the labor supply shock being real and largely unoffset in the first post-termination decade.⁹

5 Results

5.1 Child Labor Supply

Table 2 presents the main difference-in-differences estimates for children aged 14-17 in farm households across three labor supply outcomes. The results consistently point in the same direction: Bracero termination increased child farm labor participation in states more dependent on Bracero workers prior to 1964.

Panel A reports results for the farm laborer indicator. Column (2), which includes the full set of individual controls, shows that moving from zero to complete pre-termination Bracero dependence increased the probability of a farm child aged 14-17 being classified as a farm laborer by approximately 14.6 percentage points ($p = 0.040$). The no-controls estimate in column (1) is nearly identical at 15.1 percentage points ($p = 0.050$), indicating that the result is not driven by individual-level controls and that treatment is orthogonal to observable characteristics. Column (3), which restricts to the 1960-70 comparison window immediately surrounding the termination, yields a coefficient of 11.5 percentage points ($p = 0.042$), which confirms that the effect is present even in the cleanest identification window and is not an artifact of the more distant 1980 post-period. To contextualize the baseline magnitude, the farm laborer rate among farm children aged 14-17 in 1960 was approximately 24%. A 14.6 percentage point increase therefore represents an increase of roughly two-thirds relative to the pre-termination baseline for a state moving from

⁹Non-Mexican foreign seasonal farm workers, including Jamaican, Bahamian, and Canadian workers, did not increase meaningfully following Bracero termination, confirming that the labor supply shock was not offset through other foreign worker channels (see Figure A.1).

zero to complete Bracero dependence. For illustrative purposes, a state like Texas with a pre-termination Bracero share of 24.1% would experience an increase of approximately 3.5 percentage points, while a state like New Mexico with a share of 60.9% would experience an increase of approximately 8.9 percentage points. As Bracero workers were employed seasonally and the Census captures labor supply in a single reference week, these estimates likely understate the total increase in child farm labor over the full agricultural year and should be interpreted as lower bounds on the true labor supply response.

Panel B repeats the analysis using an indicator for any positive work hours in the prior week as the outcome. The probability of reporting any positive work hours increased by approximately 19.8 percentage points without controls ($p = 0.013$) and 18.9 percentage points with controls ($p = 0.018$). The 1960-70 only estimate is 16.1 percentage points and is the most precisely estimated result in the table ($p = 0.005$), reinforcing that the effect is concentrated in the immediate post-termination period. The any-work-hours binary captures a broader set of working children than the farm laborer indicator, including children who report some work hours but whose primary occupation is not classified as farm labor. The larger magnitude on this outcome suggests that Bracero termination induced not only a shift into formal farm labor status but also an increase in part-time work participation among children who remained primarily in school, consistent with farm families deploying children in farm tasks outside of school hours rather than withdrawing them from school entirely.

Panel C reports results for hours worked in levels. The point estimate of 5.700 additional hours per week with controls is positive and marginally significant ($p = 0.066$), and the 1960-70 estimate of 4.560 hours is significant at the five percent level ($p = 0.021$). These estimates reflect the unconditional effect on all farm children including the roughly 80% who report zero hours worked, so the implied effect on children who actually enter farm work is substantially larger. Furthermore, because the Census is conducted in spring rather than during peak harvest months, the hours worked measure likely understates total annual

child farm labor and the work-school tradeoff occurring during summer and fall harvest seasons, reinforcing the interpretation of all estimates as lower bounds on the true labor supply response. Across all three panels the direction of the effect is consistent, and the 1960-70 window produces results that are at least as strong as the full sample, providing reassurance that the findings are not driven by the more distant pre- and post-periods.

Figure 3 presents event study estimates for all three child labor outcomes among children aged 14-17. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero across all panels, supporting the parallel trends assumption; the null 1940 coefficient is particularly reassuring as it predates the Bracero program entirely, suggesting that high and low Bracero states did not have meaningfully different child labor trajectories before the program began. Joint F-tests of the 1940 and 1950 pre-trend coefficients confirm this: p-values are 0.29, 0.21, and 0.39 for the farm laborer indicator, any-work-hours binary, and hours worked, respectively, failing to reject parallel trends across all three outcomes.

While the pre-termination point estimates are negative in sign, this pattern is directionally consistent with the historical trajectory of the Bracero program itself: as Mexican seasonal labor expanded through the 1940s and 1950s, high-exposure states were actively substituting immigrant workers for family labor, plausibly generating a modest pre-termination decline in child farm labor relative to low-exposure states. If anything, this pre-existing negative differential implies that the post-termination labor supply response estimated here understates the true effect of immigrant labor scarcity, since the counterfactual trend in high-Bracero states would have continued downward absent termination.

The 1970 coefficients are positive across all three outcomes, with the any-work-hours binary significant at the five percent level and the farm laborer and hours worked outcomes significant at the ten percent level, indicating that the child labor response materialized immediately in the first post-termination Census year. By 1980, coefficients attenuate toward zero across all outcomes, plausibly reflecting both the expansion of unauthorized Mexican migration during the 1970s, which partially restored foreign labor access, and the

accelerated mechanization of labor-intensive crops documented by San (2023).

Figure A.4 presents corroborating event study estimates restricting hours worked to children classified as farm laborers, with hours set to zero for all others. Both the hours level and binary indicator yield a consistent pattern, which suggests that the main results are not sensitive to how farm labor hours are measured. Given that the majority of children report zero farm labor hours, the unconditional estimates are naturally attenuated and should be interpreted as lower bounds on the intensive margin response among children who actually work.

Overall, these findings are consistent with evidence from more recent immigration enforcement episodes. Luo et al. (2018) find that Arizona's Legal Arizona Workers Act, which reduced the local undocumented immigrant population, increased the probability of farm family labor participation by 3-5.5 percentage points, with even larger effects among non-Hispanic farm households, broadly comparable to the state-level implied effects from the estimates once treatment scaling is accounted for. The parallel between their findings and ours suggests that farm household labor supply responses to immigrant worker scarcity are not specific to the mid-twentieth century Bracero context but reflect a more general mechanism linking immigration restrictions to within-household labor reallocation in agriculture.

5.2 Schooling and Grade Attainment

Having established that Bracero termination increased child farm labor participation, I now turn to whether this additional work came at a cost to children's education. The welfare implications of the labor supply response documented above depend critically on whether increased farm work displaced schooling or was absorbed around it.

Table 3 presents results for educational outcomes across four measures: school attendance during the reference period prior to the Census, highest grade attended, and grade-for-age. Moving from zero to complete pre-termination Bracero dependence decreased

the probability of school attendance by 13.4 percentage points ($p < 0.10$), a statistically significant reduction at the ten percent level and the strongest result in the table. This suggests that Bracero termination reduced the probability of school attendance among farm children in high-exposure states. For highest grade attended among all children aged 14-17, the estimate reflects a decrease of 0.525 years of schooling ($p = 0.424$), statistically indistinguishable from zero. The absence of any attainment effect is notable given the significant attendance result: it suggests that children who left school may have done so only temporarily or at the margin of attendance rather than dropping out in ways that reduced their completed grades. Finally, grade-for-age decreased by 5.9 percentage points ($p = 0.563$), statistically indistinguishable from zero and small in magnitude, indicating that Bracero termination did not cause children to fall behind expected grade levels. The 1960-70 window estimates are similarly null across all three schooling outcomes, confirming that the patterns documented here are not artifacts of the more distant pre- and post-periods.

Figure 4 presents event study estimates for the three schooling outcomes. Across all three panels, pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. Joint F-tests of the pre-trend coefficients yield p-values of 0.34, 0.20, and 0.67 for school attendance, highest grade attended, and grade-for-age, confirming no statistically meaningful differential pre-trends across high and low Bracero states. For school attendance in panel (a), the post-termination coefficients are negative in both 1970 and 1980, consistent with the significant table estimate, though the confidence intervals are wide. Panels (b) and (c), for grade-for-age and highest grade attended respectively, show post-termination coefficients that cluster near zero in both post-periods, which is consistent with the null table estimates and confirming that Bracero termination did not meaningfully affect grade progression or terminal attainment.

The attendance result, while nominally significant, is not corroborated by any of the attainment or progression outcomes. The Census attendance question covers the period from roughly February 1 onward, outside the peak summer and fall harvest season, suggesting

that the attendance decline does not simply reflect children missing school during crop harvests. A more plausible interpretation is that Bracero termination increased children's involvement in the broader set of farm tasks that persist year-round, pulling them out of school at the margins during the non-harvest season as well. That this attendance disruption did not translate into grade progression losses suggests the absences were frequent enough to register in the Census measure but not severe enough to cause grade retention or dropout. That said, the Census schooling measures do not capture absenteeism, academic performance, or cognitive development, and a child who misses scattered school days across the year but remains attending and passes to the next grade would appear unaffected in the data despite potentially meaningful learning losses.

These results suggest that the welfare cost of child labor substitution following Bracero termination was limited, at least as measured by the outcomes available in the decennial Census. The increased farm work documented in Table 2 appears to have been absorbed with at most modest disruption to formal schooling. This pattern echoes Dammert (2008), who finds that an agricultural shock in Peru increased child labor without affecting overall schooling outcomes, suggesting that agricultural disruptions can increase child labor without measurable schooling losses in contexts beyond the developing countries where this pattern has primarily been documented.

5.3 Heterogeneity by Gender

Table 4 presents split-sample estimates by gender for children aged 14-17 in farm households. The results reveal an asymmetric pattern across labor supply and schooling outcomes that is consistent with the gendered division of labor on mid-twentieth century family farms.

Panel A reports labor supply results separately for boys and girls. Among boys, Bracero termination increased the probability of being classified as a farm laborer by 24.8 percentage points ($p = 0.029$) and the probability of reporting any positive work hours by 29.0

percentage points ($p = 0.019$). Hours worked per week increased by approximately 8.732 hours, though this estimate is marginally significant ($p = 0.090$). To contextualize these magnitudes, the farm laborer rate among boys aged 14-17 in farm households in 1960 was substantially higher than for girls, reflecting the historical concentration of physically demanding crop tasks among male family members. A 24.8 percentage point increase therefore represents a substantial expansion of farm labor participation for a state moving from zero to complete Bracero dependence.

Among girls, the pattern is more muted. The farm laborer indicator is small and statistically indistinguishable from zero ($p = 0.434$), indicating that girls were not reclassified into formal farm labor occupations following termination. However, hours worked increased by 2.400 hours per week ($p = 0.019$) and the any-work-hours binary is significant at the five percent level ($p = 0.035$), suggesting that girls were drawn into informal farm tasks that do not register in occupation classifications but nonetheless represent a real increase in labor supply. The divergence between the farm laborer indicator and the hours-based outcomes for girls points to an increase in unclassified or ancillary farm work or domestic tasks that free adult operators for field work, rather than direct substitution into the crop labor roles that Bracero workers had performed.

The event study estimates for boys in Figure A.5 show pre-termination coefficients statistically indistinguishable from zero across all three labor supply outcomes, supporting the parallel trends assumption. The 1970 coefficients are positive for farm laborer and any-work-hours, consistent with the significant table estimates, and attenuate toward zero by 1980. For girls in Figure A.6, pre-trends are flat for the farm laborer indicator and any-work-hours binary, and the hours worked outcome shows a positive post-termination response in 1970, consistent with the table estimates.

Figure A.9 presents complementary event study estimates restricting hours to children classified as farm laborers. For boys, the any-work-hours binary in Panel (b) shows a positive and precisely estimated 1970 coefficient, consistent with the main gender results,

while the hours level in Panel (a) is positive but imprecisely estimated. For girls, both outcomes are near zero in the post-period, consistent with the finding that girls were drawn into informal farm tasks that do not register in occupation classifications. The negative 1950 pre-trend coefficient in Panel (c) warrants caution in interpreting the hours-based results for girls specifically.

Panel B reports schooling results by gender. Among boys, school attendance declined by 16.9 percentage points ($p = 0.032$), a meaningful and statistically significant reduction that drives the aggregate attendance result in Table 3. Highest grade attained and grade-for-age are negative in point estimate for boys but statistically indistinguishable from zero, suggesting that the attendance decline did not translate into measurable grade progression losses, at least as captured by the decennial Census. Among girls, all three schooling outcomes are small and statistically insignificant, indicating that the increased labor burden documented in Panel A did not come at a detectable human capital cost. The absence of schooling effects among girls is consistent with the informal and intermittent nature of the tasks they were deployed in, which may not have accumulated into absences substantial enough to register in the Census attendance measure or affect grade progression.

The event study estimates for boys' schooling outcomes in Figure A.7 show a negative post-termination pattern for school attendance in both 1970 and 1980, supporting the significant table estimate, while grade-for-age and highest grade attended cluster near zero in the post-period. For girls in Figure A.8, school attendance and grade-for-age show flat pre-trends and post-period coefficients near zero, reflecting the null table estimates.

Taken together, the gender results tell a coherent story about within-household labor allocation following Bracero termination. Boys substituted directly into the field labor roles vacated by departing Bracero workers and bore a measurable schooling cost as a result. Girls contributed additional labor through informal channels without visible disruption to their schooling, consistent with the gendered division of tasks in mid-twentieth century U.S. crop agriculture.

5.4 Age-Specific Effects

If Bracero termination induced farm households to deploy children as substitutes for adult migrant workers, the response should be concentrated among older children who are physically capable of performing the demanding field tasks that Bracero workers predominantly performed, including harvesting, planting, and other intensive seasonal manual labor. Table A.3 examines whether effects vary systematically across single years of age within the 14-17 range.

The results confirm this prediction. Farm laborer effects increase monotonically from 10.2 percentage points at age 14 to 18.1 percentage points at age 16, before settling at 16.6 percentage points at age 17. The any-work-hours binary shows an even sharper age gradient, rising from an insignificant 9.6 percentage points at age 14 to a precisely estimated 26.0 percentage points at age 17. Hours worked follow the same pattern, with the effect at age 14 marginally significant at 3.287 additional hours per week and the largest effect among 17-year-olds at 8.998 additional hours per week ($p = 0.048$). This monotonic increase in labor supply effects with age points to physical capacity and productivity in farm work driving the substitution of child labor for adult Bracero workers, rather than a general labor market response that would be expected to operate more uniformly across the age distribution.

The schooling results display a complementary pattern. School attendance effects are statistically indistinguishable from zero at ages 14, 15, and 16, but decreased by 20.9 percentage points among 17-year-olds. This concentration of schooling disruption at the oldest age group is consistent with 17-year-olds facing the sharpest school-work tradeoff, as they are at the margin of completing or abandoning secondary schooling entirely. Younger children, by contrast, appear to have remained attending school even as their labor supply increased, suggesting their involvement was light or intermittent enough not to accumulate into absences substantial enough to register in the Census attendance measure. Grade progression outcomes remain statistically indistinguishable from zero across all four age

groups, confirming that even the attendance decline among 17-year-olds did not translate into measurable losses in grade-for-age progression or highest grade attained. Together, the age profile of effects supports a farm labor substitution interpretation and suggests that the welfare costs of Bracero termination for farm children were borne disproportionately by older teenagers who were productive enough to substitute meaningfully for adult workers but old enough to face genuine tradeoffs between farm work and schooling completion.

5.5 Sample Composition and Falsification Tests

5.5.1 Selection into Farm Households

Because the main analysis is restricted to children residing in farm households, the estimates could be biased if the termination of the Bracero program differentially altered which households remained in farming across states with different levels of pre-termination Bracero reliance. For example, if more heavily exposed states experienced greater exit from farming after 1964, the composition of the observed farm-household sample could change even in the absence of any effect on child labor. To assess this potential source of sample selection, I estimate the baseline difference-in-differences specification using an indicator for residing in a farm household as the outcome, defined over the full sample of U.S.-born children aged 14-17 rather than restricting the sample to farm households.

Table A.4 reports the results. The estimated coefficient is small in magnitude and statistically indistinguishable from zero across all three specifications. Figure ?? presents the corresponding event-study estimates. The pre-termination coefficients for 1940 and 1950 are both statistically indistinguishable from zero, providing no evidence of differential pre-treatment trends in farm household status across states with different levels of Bracero reliance. Likewise, the post-termination coefficients for 1970 and 1980 remain statistically indistinguishable from zero, indicating that changes in the likelihood of residing in a farm household were not systematically related to pre-termination Bracero exposure at any point following the program's termination.

These results suggest that differential entry into or exit from farm households is unlikely to explain the main findings. The absence of systematic changes in farm household status across states with different levels of Bracero reliance supports the interpretation that the estimated effects reflect changes in child labor behavior within farm households rather than compositional changes in the analysis sample.

5.5.2 Falsification Test

As a falsification test, I estimate equation 1 for children aged 14-17 in non-farm households. If Bracero termination affected child labor outcomes through channels other than farm labor demand, for example through local economic spillovers or state-level shocks correlated with Bracero reliance, one would expect to find similar effects among non-farm children. Table 5 presents these results across both labor supply and schooling outcomes. In Panel A, the treatment coefficient is small, negative, and statistically indistinguishable from zero for the farm laborer indicator and any-work-hours binary, and similarly null for hours worked in levels. In Panel B, school attendance, highest grade attended, and grade-for-age are all small and statistically indistinguishable from zero. There is no evidence that Bracero termination affected either labor supply or schooling outcomes among non-farm children in high-exposure states.

These null falsification results strengthen the causal interpretation of the main findings in two ways. First, they suggest that the child labor and schooling responses documented among farm children are more consistent with the farm labor supply channel than with a general state-level trend affecting all children in high-Bracero states. Second, they make it less likely that the main results are driven by broad economic conditions or demographic changes correlated with pre-termination Bracero reliance, since such confounds would be expected to affect non-farm children in the same states as well.

5.6 Robustness Checks

5.6.1 Alternative Treatment Definitions

Table 6 examines sensitivity to alternative definitions of the treatment variable. Column (1) reproduces the baseline result using the continuous 1955 Bracero share. Column (2) replaces this with a binary indicator equal to one if the state's 1955 share exceeds 0.20. Column (3) uses the average Bracero share across 1954-63, which is more robust to anticipatory adjustments as political pressure to end the program intensified in the early 1960s. The farm laborer indicator and any-work-hours binary remain positive and statistically significant across all three columns, with magnitudes broadly comparable to the baseline. Hours worked is positive throughout but loses significance under the binary treatment definition, unsurprising given the coarser variation that definition exploits.

The schooling results are equally stable. School attendance remains negative across all three columns, with the binary and baseline estimates significant at the ten percent level and the averaged exposure measure producing a similarly sized but statistically insignificant coefficient. Highest grade attended and grade-for-age remain statistically indistinguishable from zero across all specifications, providing no evidence that Bracero termination translated into lasting losses in educational attainment regardless of how pre-termination exposure is measured.

5.6.2 Alternative Specifications

Table A.5 examines robustness to alternative specifications across all six outcomes. Column (1) reproduces the baseline estimates. Column (2) replaces the continuous age polynomial with age fixed effects, which nonparametrically absorb any age composition differences across treatment states. The results are virtually identical to the baseline, confirming that the age polynomial adequately captures the within-sample age gradient. Column (3) drops household head controls, and coefficients remain nearly unchanged. Column (4) drops

California and Texas, the two states with the largest absolute Bracero worker counts. Labor supply results remain positive and significant, though the hours-worked estimate loses significance, consistent with these states contributing meaningfully to estimation precision. Schooling estimates remain statistically insignificant throughout, and the binary labor supply estimates are stable in both magnitude and significance across all four specifications. Overall, the results are robust to alternative functional form assumptions, control sets, and sample restrictions.

6 Conclusion

This paper provides evidence that the termination of the Bracero program in 1964 increased child farm labor participation among U.S. farm households. Using cross-state variation in pre-termination Bracero reliance and decennial Census microdata, I find that farm children aged 14-17 in states more exposed to the program's end were significantly more likely to be employed as farm laborers and to report positive work hours in the post-termination period. These effects are robust across multiple outcome measures, alternative treatment definitions, and alternative specifications, and are supported by event study evidence of clean pre-trends. The labor supply response was concentrated among male children and among older teenagers, consistent with the gendered division of labor on family farms and with physical capacity driving substitution into the demanding field tasks that Bracero workers had performed. Girls contributed additional labor through informal channels that did not register in occupation classifications, and 17-year-olds bore a measurable school attendance cost that younger children did not. Falsification tests on non-farm households make it less likely that the results are driven by broad state-level economic trends correlated with Bracero exposure.

The primary estimates reflect medium-run effects, as the first post-termination Census year is 1970, five to six years after the program's end. The effects measured here therefore

capture the farm household labor supply response after an initial adjustment period rather than the immediate shock. Educational attainment and grade progression are broadly unaffected, suggesting the additional farm work was frequent enough to affect attendance at the margins but not severe enough to cause grade retention or dropout at measurable rates. The exception is school attendance among boys and among 17-year-olds, where the estimates are negative and statistically significant, indicating that the welfare costs of Bracero termination were real but concentrated among the oldest and most productive child workers rather than spread uniformly across the farm child population.

These findings carry several implications for contemporary policy. First, the results demonstrate that the FLSA agricultural exemption functioned as a meaningful margin of adjustment, with farm households increasing child labor supply in response to adult labor scarcity. Fan et al. (2014) document that child agricultural labor remains substantial in the modern United States under the same exemption, and the present results suggest that this exemption can operate as a meaningful behavioral margin rather than merely a dormant legal distinction. Policymakers considering reform of the FLSA agricultural exemption should account for the possibility that tightening it in the absence of adequate adult labor supply could generate unintended consequences for farm household welfare.

Second, the results raise questions relevant to ongoing debates over the H-2A agricultural guest worker program, which currently supplies authorized seasonal farm labor in a manner structurally similar to the Bracero program. Whether the household labor reallocation mechanism documented here would operate with similar force under contemporary H-2A constraints is an open empirical question, as farms today face a different mechanization frontier and H-2A restrictions tend to be more gradual and anticipated than the abrupt 1964 termination studied here. Nonetheless, the findings suggest that H-2A availability may relieve pressure on family labor within farm households, and that restrictions on H-2A access could increase child labor burdens in ways that do not register in standard administrative employment and wage data, a possibility worth investigating

with more contemporary data as H-2A policy continues to evolve.

Third, more broadly, the paper shows that immigration enforcement in agricultural labor markets can reshape labor allocation within native farm households in ways that may not be fully captured in conventional labor market statistics but can nevertheless have important implications for children. The schooling results provide some reassurance that these costs did not extend to lasting human capital losses in the historical context studied here, though the attendance effects among boys and older teenagers suggest the welfare consequences were not uniformly benign and warrant attention in evaluating the full distributional impact of agricultural immigration policy.

References

- Ali, U., Brown, J. H., and Herbst, C. M. (2024). Secure communities as immigration enforcement: How secure is the child care market? *Journal of Public Economics*, 233:105101.
- Amuedo-Dorantes, C. and Antman, F. (2017). Schooling and labor market effects of temporary authorization: Evidence from DACA. *Journal of Population Economics*, 30(1):339–373.
- Amuedo-Dorantes, C., Arenas-Arroyo, E., and Sevilla, A. (2018). Immigration enforcement and economic resources of children with likely unauthorized parents. *Journal of Public Economics*, 158:63–78.
- Baker, R. B. (2015). From the field to the classroom: the Boll Weevil’s impact on education in rural Georgia. *The Journal of Economic History*, 75(4):1128–1160.
- Baker, R. B., Blanchette, J., and Eriksson, K. (2020). Long-run impacts of agricultural shocks on educational attainment: Evidence from the Boll Weevil. *The Journal of Economic History*, 80(1):136–174.
- Beegle, K., Dehejia, R. H., and Gatti, R. (2006). Child labor and agricultural shocks. *Journal of Development Economics*, 81(1):80–96.
- Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of Economic Studies*, 89(1):181–213.
- Boustan, L. P. (2010). Was postwar suburbanization “white flight”? evidence from the black migration. *The Quarterly Journal of Economics*, 125(1):417–443.
- Calavita, K. (2010). *Inside the state: The Bracero Program, immigration, and the INS*. Quid Pro Books.
- Charlton, D. and Kostandini, G. (2021). Can technology compensate for a labor shortage? effects of 287 (g) immigration policies on the us dairy industry. *American Journal of Agricultural Economics*, 103(1):70–89.

- Clemens, M. A., Lewis, E. G., and Postel, H. M. (2018). Immigration restrictions as active labor market policy: Evidence from the Mexican Bracero exclusion. *American Economic Review*, 108(6):1468–1487.
- Colmer, J. (2021). Rainfall variability, child labor, and human capital accumulation in rural ethiopia. *American Journal of Agricultural Economics*, 103(3):858–877.
- Dammert, A. C. (2008). Child labor and schooling response to changes in coca production in rural peru. *Journal of Development Economics*, 86(1):164–180.
- East, C. N. (2020). The effect of food stamps on children’s health: Evidence from immigrants’ changing eligibility. *Journal of Human Resources*, 55(2):387–427.
- Fan, M., Huston, M., and Pena, A. (2014). Determinants of child labor in the modern United States: Evidence from agricultural workers and their children and concerns for ongoing public policy. *Economics Bulletin*, 36(1):125–145.
- Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.
- Hainmueller, J., Lawrence, D., Martén, L., Black, B., Figueroa, L., Hotard, M., and Laitin, D. D. (2017). Protecting unauthorized immigrant mothers improves their children’s mental health. *Science*, 357(6355):1041–1044.
- Ifft, J. and Jodlowski, M. (2022). Is ICE freezing US agriculture? farm-level adjustment to increased local immigration enforcement. *Labour Economics*, 78:102203.
- Kostandini, G., Mykerezzi, E., and Escalante, C. (2014). The impact of immigration enforcement on the US farming sector. *American Journal of Agricultural Economics*, 96(1):172–192.
- Kruger, D. I. (2007). Coffee production effects on child labor and schooling in rural brazil. *Journal of Development Economics*, 82(2):448–463.

- Liu, J.-Y. (2026). Natural disaster and intergenerational occupational choice in agriculture: Evidence from the US Dust Bowl. *American Journal of Agricultural Economics*.
- Luo, T. and Kostandini, G. (2022). Stringent immigration enforcement and responses of the immigrant-intensive sector: Evidence from E-Verify adoption in Arizona. *American Journal of Agricultural Economics*, 104(4):1411–1434.
- Luo, T., Kostandini, G., and Jordan, J. L. (2018). The impact of lawa on the family labour supply among farm households. *European Review of Agricultural Economics*, 45(5):857–878.
- Manacorda, M. (2006). Child labor and the labor supply of other household members: Evidence from 1920 America. *American Economic Review*, 96(5):1788–1801.
- Martinez-Donate, A., Tellez Lieberman, J., Bakely, L., Correa, C., Valdez, C., McGhee Hassrick, E., and Rangel Gomez, G. (2020). Deporting immigrant parents: Impact on the health and well-being of their citizen children. *European Journal of Public Health*, 30:165–203.
- Moehling, C. M. (1999). State child labor laws and the decline of child labor. *Explorations in Economic History*, 36(1):72–106.
- Monras, J. (2020). Immigration and wage dynamics: Evidence from the Mexican peso crisis. *Journal of Political Economy*, 128(8):3017–3089.
- Richards, T. J. (2018). Immigration reform and farm labor markets. *American Journal of Agricultural Economics*, 100(4):1050–1071.
- Ruggles, S., Flood, S., Sobek, M., Backman, D., Chen, A., Cooper, G., Richards, S., Rodgers, R., and Schouweiler, M. (2024). IPUMS USA: Version 14.0. <https://doi.org/10.18128/D010.V14.0>.
- Rutledge, Z. and Mérel, P. (2023). Farm labor supply and fruit and vegetable production. *American Journal of Agricultural Economics*, 105(2):644–673.

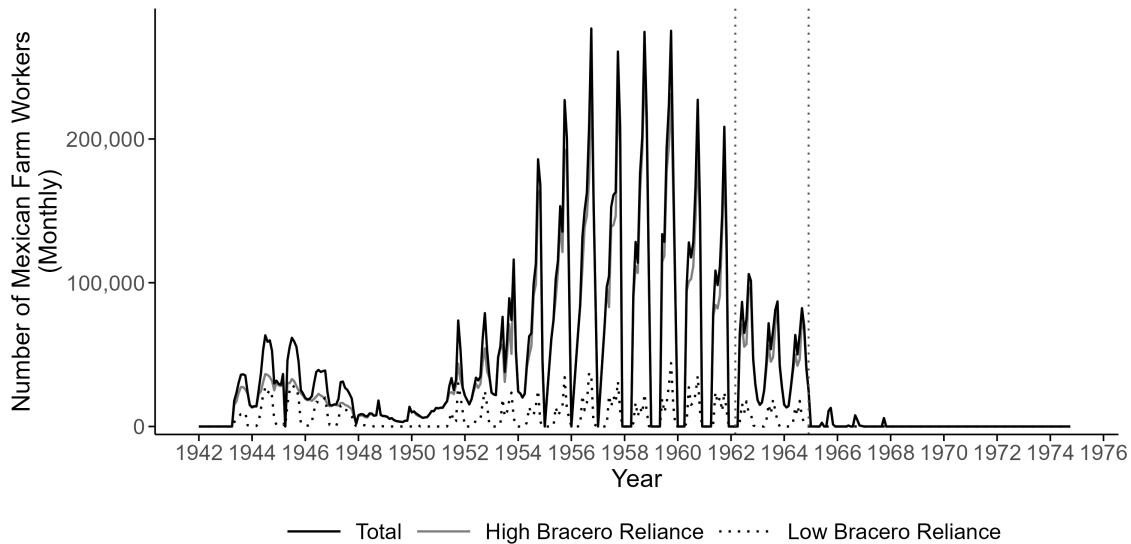
San, S. (2023). Labor supply and directed technical change: Evidence from the termination of the Bracero program in 1964. *American Economic Journal: Applied Economics*, 15(1):136–163.

Stephens Jr, M. and Yang, D.-Y. (2014). Compulsory education and the benefits of schooling. *American Economic Review*, 104(6):1777–1792.

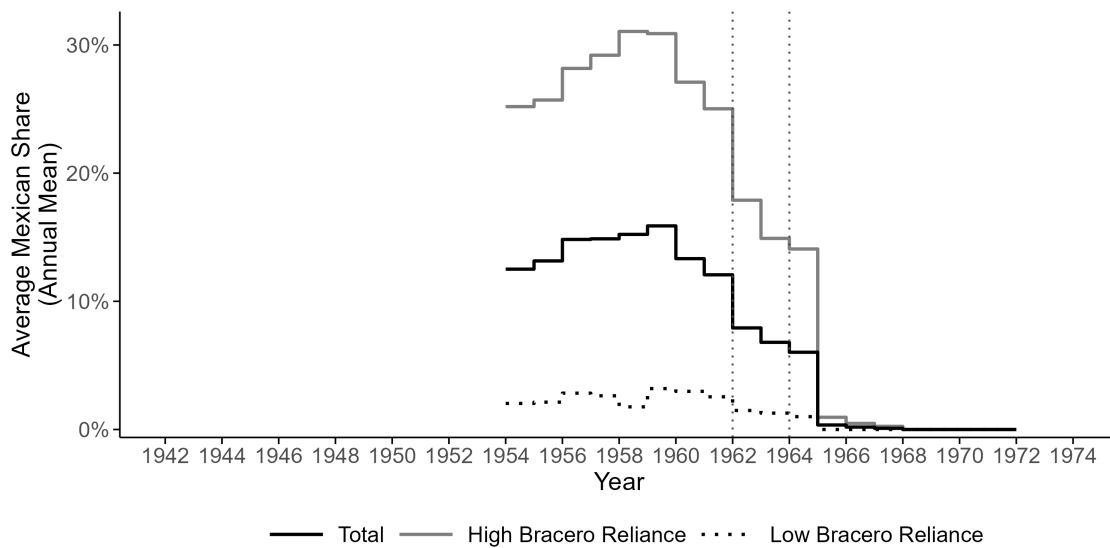
Zahniser, S., Hertz, T., Dixon, P., and Rimmer, M. (2012). Immigration policy and its possible effects on us agriculture and the market for hired farm labor: a simulation analysis. *American Journal of Agricultural Economics*, 94(2):477–482.

Figures and Tables

Figure 1: Number and Share of Mexican Farm Workers in the United States



(a) Number of Mexican farm workers in the U.S. by month



(b) Share of Mexican farm workers in the U.S. by year

Notes: Data are from archival administrative records compiled by Clemens et al. (2018). Both panels cover 1942-1974. Vertical dotted lines indicate 1962 and 1964, marking the beginning of the political debate over the program and the year of formal termination, respectively. High Bracero reliance states are defined as states where the 1955 Bracero share exceeded 0.20.

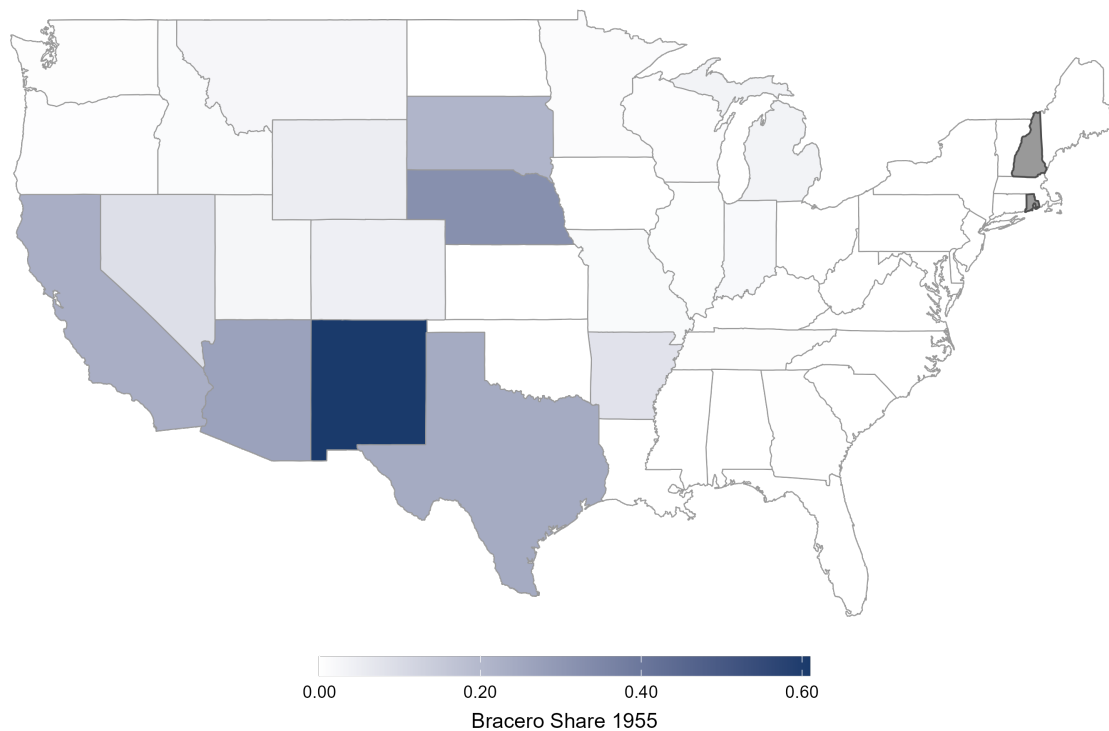


Figure 2: Pre-Termination Bracero Exposure by State (1955)

Notes: Map displays the 1955 state-level Bracero share, defined as the fraction of total hired seasonal farm workers who were Mexican (Bracero) workers in 1955. Darker shading indicates higher pre-termination Bracero reliance. States in white had no recorded Bracero workers in the archival administrative records for 1955. Data are from archival administrative records compiled by Clemens et al. (2018). Rhode Island and New Hampshire are shaded in gray because farm worker stock data are missing in the original sources.

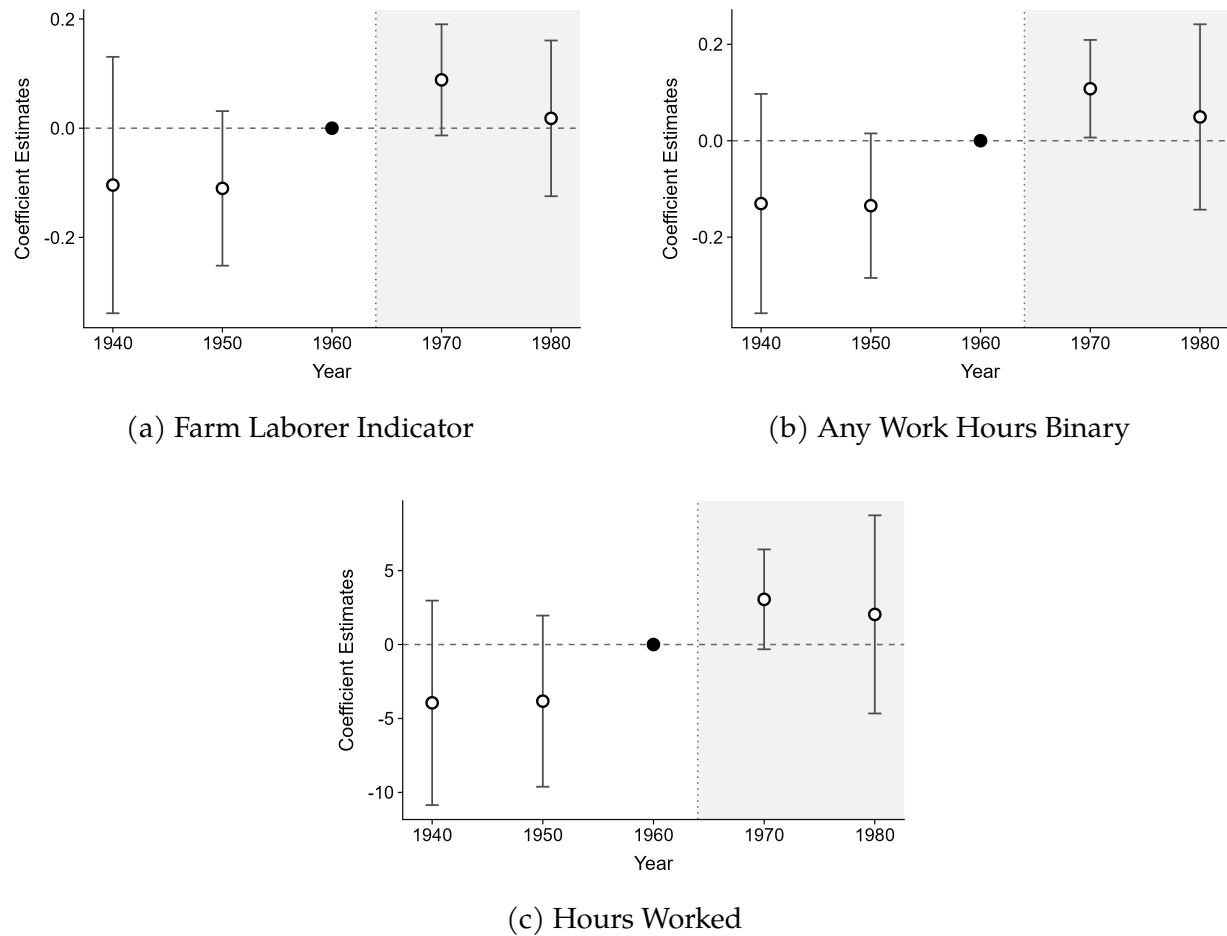
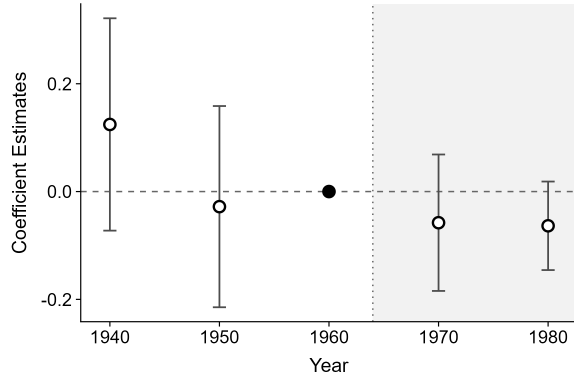
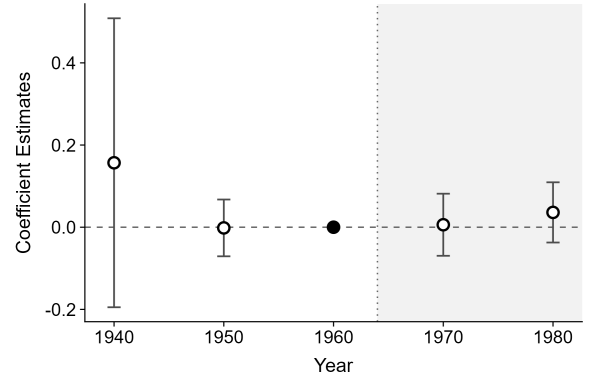


Figure 3: Event Study: Child Farm Labor Outcomes

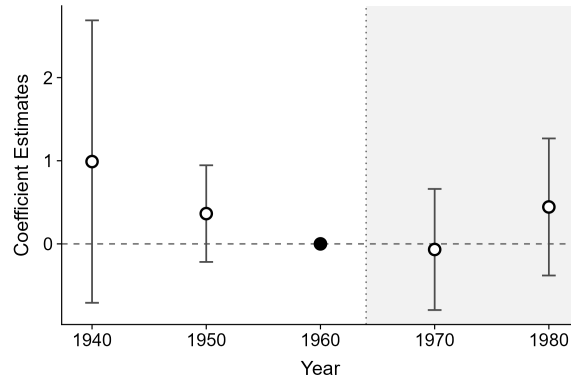
Notes: Event study estimates from equation (2) for children aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. Post-termination coefficients for 1970 and 1980 capture the dynamic response to Bracero termination. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.



(a) School Attendance



(b) Grade for Age



(c) Highest Grade Attended

Figure 4: Event Study: Schooling Outcomes

Notes: Event study estimates from equation (2) for children aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. All post-termination coefficients are statistically indistinguishable from zero, consistent with no effect of Bracero termination on schooling outcomes. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max
<i>Panel A: Farm Household Status (Ages 14-17, Full Sample)</i>				
Farm household	0.125	0.331	0	1
Observations	1,398,243			
<i>Panel B: Labor Supply Outcomes (Ages 14-17, Farm Sample)</i>				
Farm laborer indicator	0.194	0.396	0	1
Any work hours	0.201	0.401	0	1
Hours worked last week	6.68	15.711	0	65
Observations	120,926			
<i>Panel C: Schooling Outcomes (Ages 14-17, Farm Sample)</i>				
School attendance	0.798	0.402	0	1
Grade for age	0.868	0.338	0	1
Highest grade attended	11.101	2.239	1	19
Observations	82,202			
<i>Panel D: Individual Characteristics (Ages 14-17, Farm Sample)</i>				
Age	15.490	1.114	14	17
Female	0.476	0.499	0	1
Black	0.137	0.344	0	1
Latino	0.014	0.118	0	1
Family size	6.006	2.525	1	26
Rank among children under 18 in the household	1.384	0.761	1	15
Observations	120,926			

Notes: Panel A is computed for all U.S.-born children aged 14-17 across five Census years (1940, 1950, 1960, 1970, 1980). Panels B and D are restricted to farm household children, weighted using person weights scaled by 0.5 for 1970 observations to reflect the pooling of Form 1 and Form 2 state samples. Panel C restricts the 1970 sample to Form 2 only and uses original Census person weights; the smaller sample reflects missing school attendance values in the 1970 Census Form 2 sample. School attendance is coded as one if the child is currently attending school and zero otherwise.

Table 2: Effect of Bracero Termination on Child Farm Labor

	(1) No Controls	(2) With Controls	(3) 1960-1970 Only
<i>Panel A: Farm Laborer Indicator</i>			
Treatment	0.151** (0.075)	0.146** (0.069)	0.115** (0.055)
<i>Panel B: Any Work Hours (Binary)</i>			
Treatment	0.198** (0.077)	0.189** (0.077)	0.161*** (0.054)
<i>Panel C: Hours Worked (Continuous)</i>			
Treatment	5.962* (2.968)	5.700* (3.022)	4.560** (1.902)
Individual Controls	No	Yes	Yes
Household Head Controls	No	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	120,926	120,926	30,915

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children aged 14-17 in farm households across five Census years (1940, 1950, 1960, 1970, 1980). Column (3) restricts the sample to the 1960 and 1970 Census years only. The treatment variable is the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). For 1970, both Form 1 and Form 2 state samples are pooled for labor supply outcomes, with person weights scaled by 0.5 to reflect the combined 2% sampling rate. Controls include family size, age, age squared, female indicator, race/ethnicity indicators, and birth order. Head controls include household head age, sex, marital status, and highest grade attended. Panel A outcome is a farm labor indicator equal to one if the child's occupation is coded as farm laborer, wage worker, or unpaid family farm worker. Panel B outcome is an indicator equal to one if the child reports positive work hours in the reference week. Panel C outcome is hours worked last week, constructed by assigning bracket midpoints to the Census hours categories. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Effect of Bracero Termination on Schooling Outcomes

	(1) No Controls	(2) With Controls	(3) 1960-1970 Only
<i>Panel A: School Attendance</i>			
Treatment	-0.129* (0.068)	-0.134* (0.079)	-0.061 (0.060)
Observations	82,202	82,202	22,237
<i>Panel B: Grade for Age</i>			
Treatment	-0.050 (0.112)	-0.059 (0.102)	-0.030 (0.045)
Observations	120,926	120,926	30,915
<i>Panel C: Highest Grade Attended</i>			
Treatment	-0.456 (0.732)	-0.525 (0.656)	-0.276 (0.386)
Observations	120,926	120,926	30,915
Individual Controls	No	Yes	Yes
Household Head Controls	No	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children aged 14-17 in farm households across five Census years (1940, 1950, 1960, 1970, 1980). Column (3) restricts the sample to the 1960 and 1970 Census years only. The treatment variable is the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). For 1970, only Form 2 of the state sample is used for schooling outcomes, as school attendance was not asked on Form 1. Person weights are the original Census person weights for schooling outcomes. Controls include family size, age, age squared, female indicator, race/ethnicity indicators, and birth order. Head controls include household head age, sex, marital status, and highest grade attended. Panel A outcome is a school attendance indicator equal to one if the child attended school during the reference period prior to the Census. Panel B outcome is a grade-for-age index equal to one if the child is enrolled at or above the expected grade level for their age, and zero if they are behind grade level. Panel C outcome is the highest grade attended for all children aged 14-17. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Effect of Bracero Termination on Child Outcomes by Gender

	Boys			Girls		
	(1) Farm Laborer	(2) Any Work Hours	(3) Hours Worked	(4) Farm Laborer	(5) Any Work Hours	(6) Hours Worked
<i>Panel A: Labor Supply</i>						
Treatment	0.248** (0.112)	0.290** (0.122)	8.732* (5.043)	0.033 (0.042)	0.076** (0.035)	2.400** (0.988)
Observations	63,376	63,376	63,376	57,550	57,550	57,550
<i>Panel B: Schooling</i>						
	School Attendance	Highest Grade	Grade for Age	School Attendance	Highest Grade	Grade for Age
Treatment	-0.169** (0.076)	-0.547 (0.667)	-0.066 (0.112)	-0.096 (0.097)	-0.522 (0.667)	-0.055 (0.091)
Observations	42,867	63,376	63,376	39,335	57,550	57,550
Individual Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household Head Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children aged 14-17 in farm households across five Census years (1940, 1950, 1960, 1970, 1980). The treatment variable is the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). Labor supply outcomes use person weights scaled by 0.5 for 1970 observations to reflect the pooling of Form 1 and Form 2 state samples. For schooling outcomes, the 1970 sample is restricted to Form 2 only, as school attendance was not asked on Form 1, and original Census person weights are used. Controls include family size, age, age squared, female indicator, race/ethnicity indicators, and birth order. Head controls include household head age, sex, marital status, and highest grade attended. Columns (1)-(3) report estimates for boys and columns (4)-(6) report estimates for girls. Panel A outcomes are labor supply measures: columns (1) and (4) use a farm laborer indicator equal to one if the child's occupation is coded as farm laborer; columns (2) and (5) use an indicator equal to one if the child reports any positive work hours in the reference week; columns (3) and (6) use hours worked last week in levels. Panel B outcomes are schooling measures: columns (1) and (4) use a school attendance indicator equal to one if the child attended school during the reference period prior to the Census; columns (2) and (5) use highest grade attended for all children aged 14-17; columns (3) and (6) use a grade-for-age index equal to one if the child is at or above the expected grade level for their age. Lower observations for school attendance reflect missing values in the 1970 Census Form 2 sample. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Falsification Test: Non-Farm Households

<i>Panel A: Labor Supply Outcomes</i>			
	(1)	(2)	(3)
	Farm Laborer	Any Work Hours	Hours Worked
Treatment	-0.013 (0.010)	-0.023 (0.030)	0.564 (1.472)
Observations	1,277,317	1,277,317	1,277,317
<i>Panel B: Schooling Outcomes</i>			
	School Attendance	Highest Grade	Grade for Age
Treatment	-0.061 (0.072)	0.004 (0.279)	0.018 (0.060)
Observations	1,057,789	1,277,317	1,277,317
Individual Controls	Yes	Yes	Yes
Household Head Controls	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children aged 14-17 in *non-farm* households across five Census years (1940, 1950, 1960, 1970, 1980). The treatment variable is the interaction of the 1955 state-level Bracero share ($\overline{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). Labor supply outcomes use person weights scaled by 0.5 for 1970 observations to reflect the pooling of Form 1 and Form 2 state samples. For schooling outcomes, the 1970 sample is restricted to Form 2 only, as school attendance was not asked on Form 1, and original Census person weights are used. Controls include family size, age, age squared, female indicator, race/ethnicity indicators, and birth order. Head controls include household head age, sex, marital status, and highest grade attended. Panel A outcomes are labor supply measures: column (1) uses a farm laborer indicator; column (2) uses an indicator equal to one if the child reports any positive work hours in the reference week; column (3) uses hours worked last week in levels. Panel B outcomes are schooling measures: column (1) uses a school attendance indicator; column (2) uses highest grade attended for all children aged 14-17; column (3) uses a grade-for-age index equal to one if the child is at or above the expected grade level for their age. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Robustness: Alternative Treatment Definitions

	(1) Continuous 1955	(2) Binary High Reliance	(3) Average 1954-63
<i>Panel A: Farm Laborer Indicator</i>			
Treatment	0.146** (0.069)	0.037* (0.020)	0.165* (0.094)
Observations	120,926	120,926	120,926
<i>Panel B: Any Work Hours (Binary)</i>			
Treatment	0.189** (0.077)	0.051** (0.022)	0.217* (0.102)
Observations	120,926	120,926	120,926
<i>Panel C: Hours Worked (Continuous)</i>			
Treatment	5.700* (3.022)	1.480 (0.918)	7.072* (3.811)
Observations	120,926	120,926	120,926
<i>Panel D: School Attendance</i>			
Treatment	-0.137* (0.080)	-0.042* (0.022)	-0.144 (0.098)
Observations	82,202	82,202	82,202
<i>Panel E: Highest Grade Attended</i>			
Treatment	-0.443 (0.664)	-0.136 (0.172)	-0.227 (0.670)
Observations	120,926	120,926	120,926
<i>Panel F: Grade for Age</i>			
Treatment	-0.052 (0.100)	-0.013 (0.029)	-0.039 (0.114)
Observations	120,926	120,926	120,926
Individual Controls	Yes	Yes	Yes
Household Head Controls	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes

Notes: Difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children aged 14-17 in farm households across five Census years (1940, 1950, 1960, 1970, 1980). Labor supply outcomes use person weights scaled by 0.5 for 1970 observations to reflect the pooling of Form 1 and Form 2 state samples. Schooling outcomes restrict the 1970 sample to Form 2 only and use original Census person weights. Controls include household size, age, age squared, female indicator, race/ethnicity indicators, and birth rank among children under 18 in the household. Head controls include household head age, sex, marital status, and highest grade attended. Column (1) reproduces the baseline treatment variable, defined as the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). Column (2) replaces the continuous treatment with a binary indicator equal to one if the state's 1955 Bracero share exceeds 0.20, interacted with the post-1964 indicator. Column (3) replaces the 1955 share with the average Bracero share across 1954-63, interacted with the post-1964 indicator; this alternative is more robust to anticipatory adjustments as political pressure to end the program intensified in the years immediately preceding termination. Panels A-C report labor supply outcomes: farm laborer indicator, any positive work hours binary, and hours worked last week in levels. Panels D-F report schooling outcomes: school attendance, highest grade attended, and grade-for-age. The lower observation count in Panel D reflects missing school attendance values in the 1970 Census Form 2 sample. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Online Supplemental Appendix

A Additional Figures and Tables

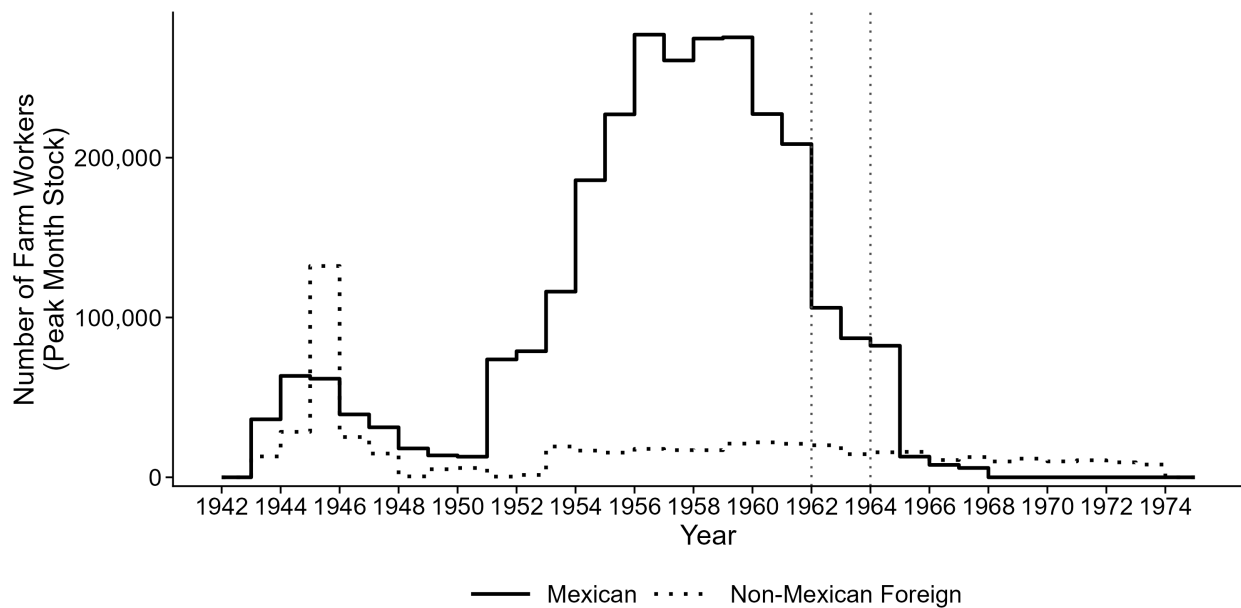


Figure A.1: Pre-Termination Bracero Workers by State (1955)

Notes: Figure plots the annual peak-month stock of Mexican and non-Mexican foreign seasonal farm workers in the United States from 1942 to 1974. Mexican workers are those admitted under the Bracero program; non-Mexican foreign workers include nationals of Jamaica, the Bahamas, the British West Indies, Canada, Puerto Rico, and other foreign countries. Vertical dotted lines indicate 1962 and 1964, marking the beginning of the political debate over the program and the year of formal termination, respectively. Data are from archival administrative records compiled by Clemens et al. (2018).

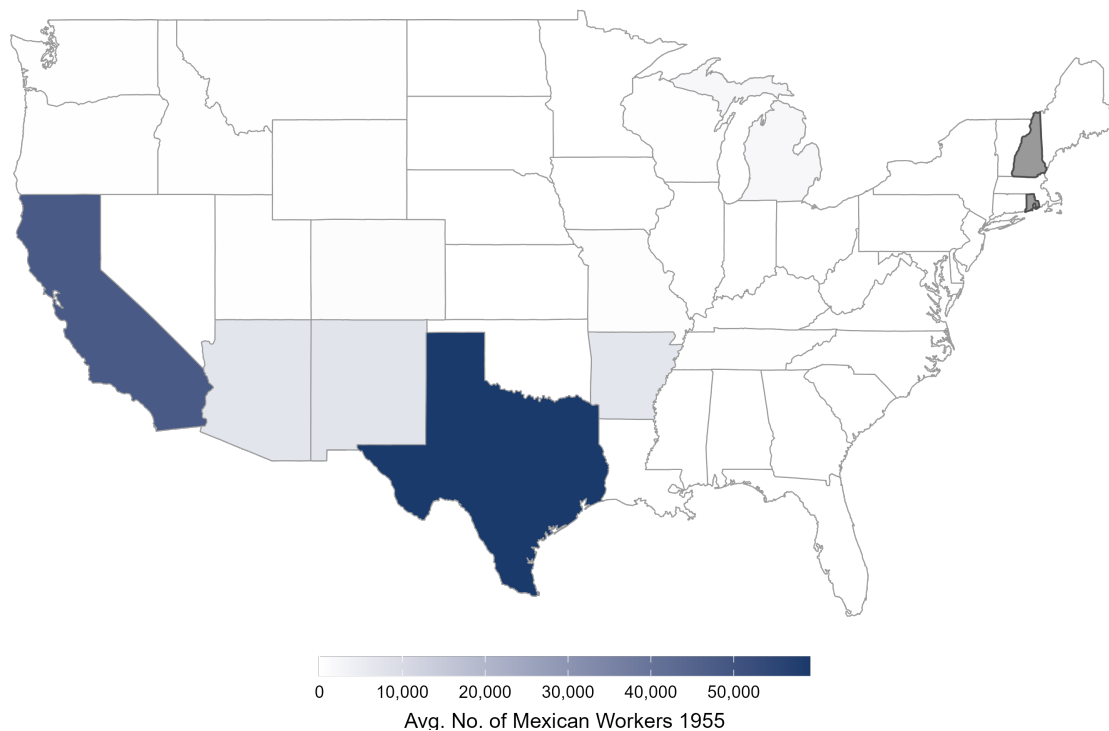


Figure A.2: Pre-Termination Bracero Workers by State (1955)

Notes: Map displays the average monthly number of Mexican Bracero workers placed in each state in 1955, averaged across available non-missing months. Darker shading indicates a larger absolute Bracero workforce. States in white had no recorded Bracero workers in 1955 administrative records. States shaded in grey (New Hampshire and Rhode Island) are excluded from the analysis due to missing farm worker stock data in the original sources. Data are from archival administrative records compiled by Clemens et al. (2018).

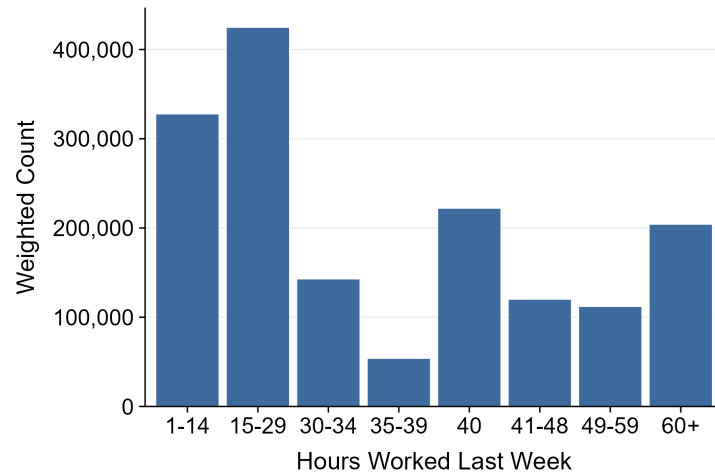
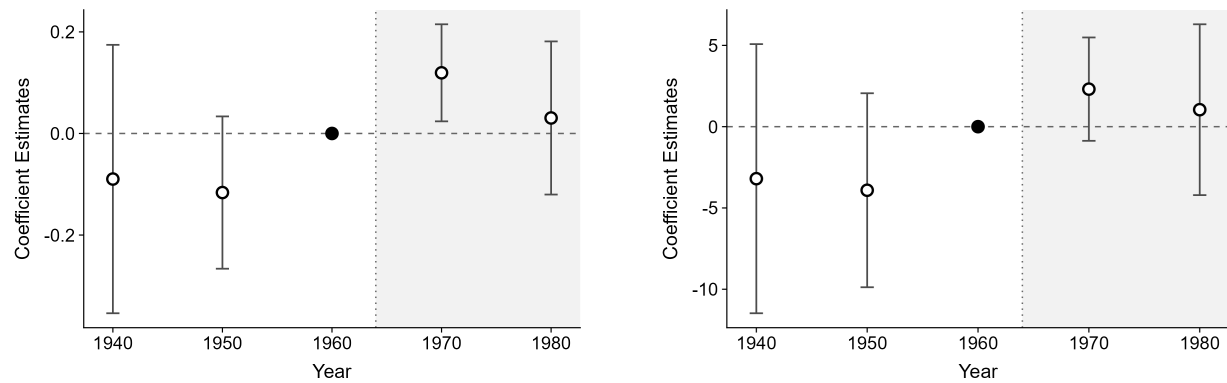


Figure A.3: Distribution of Hours Worked Among Working Farm Children

Notes: Distribution of hours worked last week among U.S.-born farm children aged 14-17 who report positive hours worked, pooled across five Census years (1940, 1950, 1960, 1970, 1980). Weighted using person weights (perwt). The dashed vertical line indicates the weighted mean. Hours are recorded in brackets in the decennial Census rather than exact values. Sample restricted to farm households.



(a) Any Work Hours on Farm Binary

(b) Hours Worked on Farm

Figure A.4: Event Study: Hours Worked on Farm

Notes: Event study estimates from equation (2) for children aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.

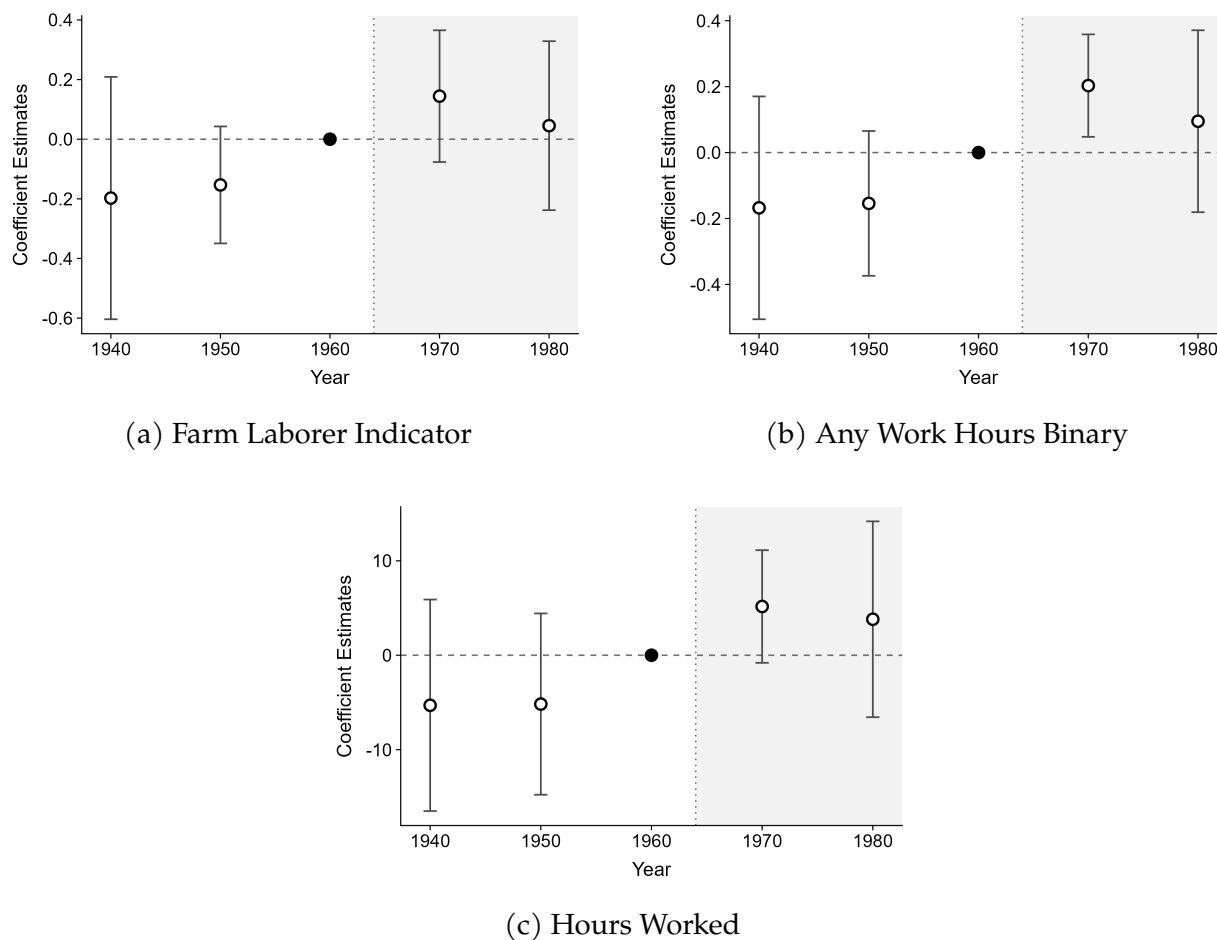


Figure A.5: Event Study: Child Farm Labor Outcomes, Boys

Notes: Event study estimates from equation (2) for boys aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. Post-termination coefficients for 1970 and 1980 capture the dynamic response to Bracero termination. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.

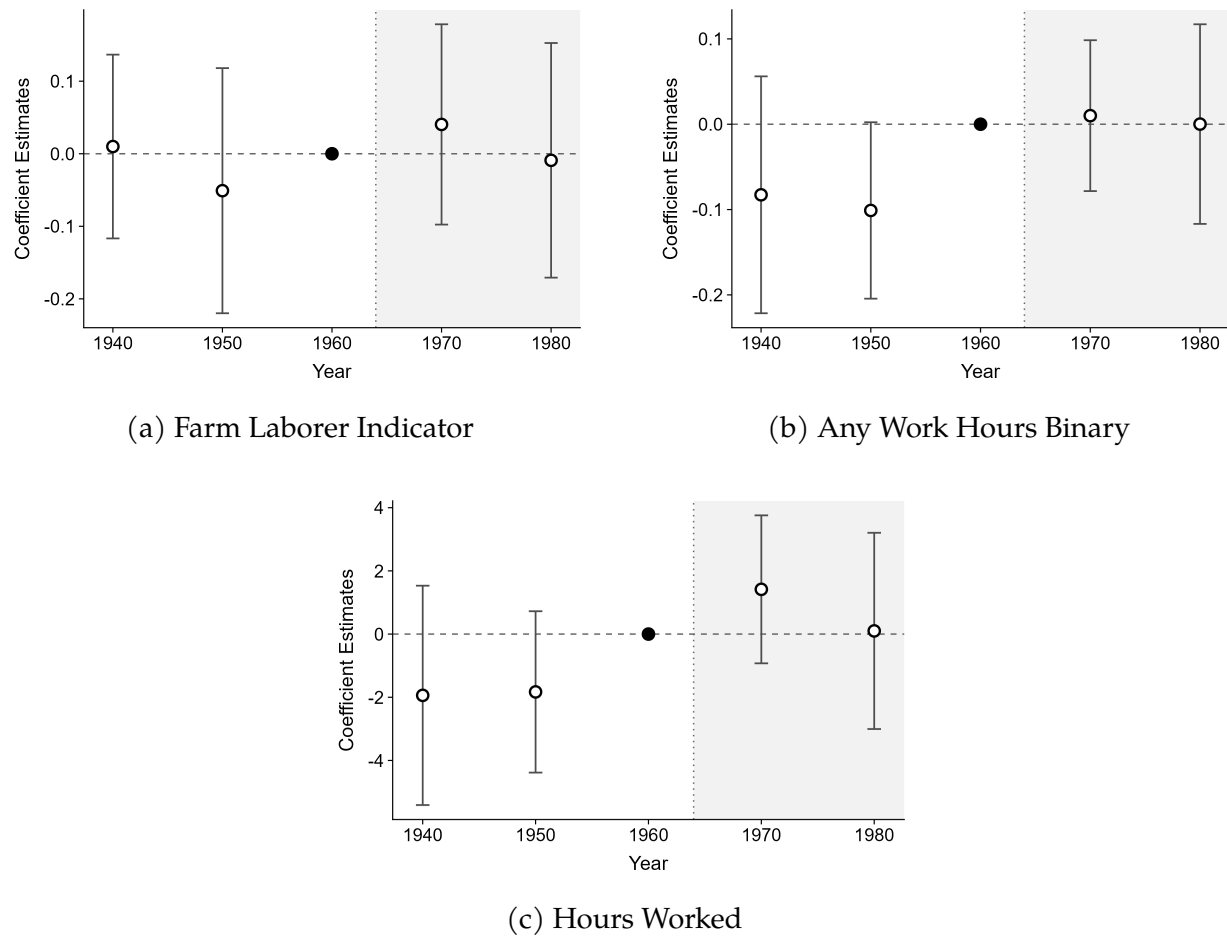
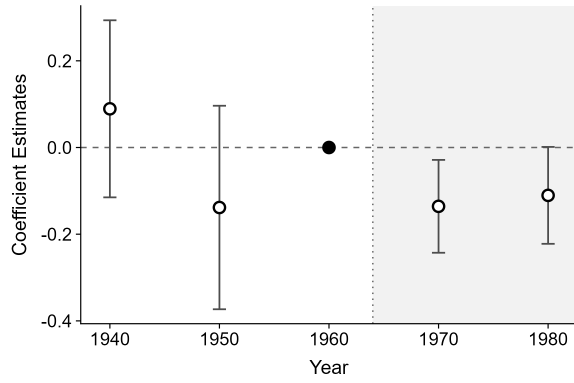
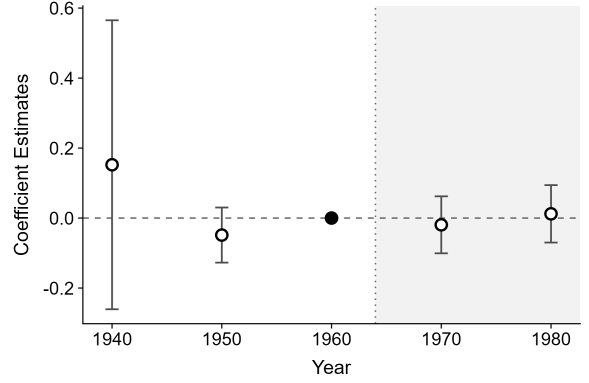


Figure A.6: Event Study: Child Farm Labor Outcomes, Girls

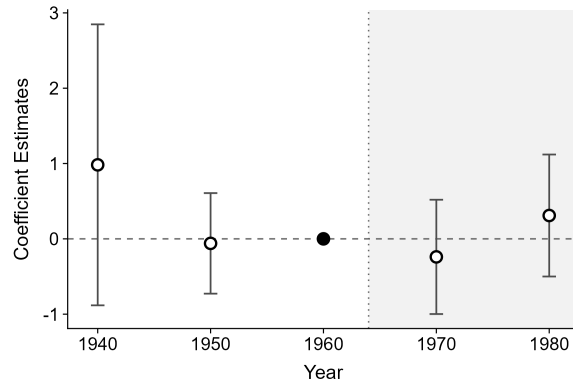
Notes: Event study estimates from equation (2) for girls aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. All post-termination coefficients are statistically indistinguishable from zero, consistent with no effect of Bracero termination on schooling outcomes. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.



(a) School Attendance



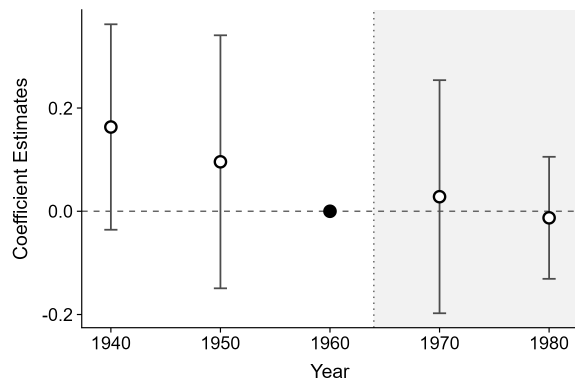
(b) Grade for Age



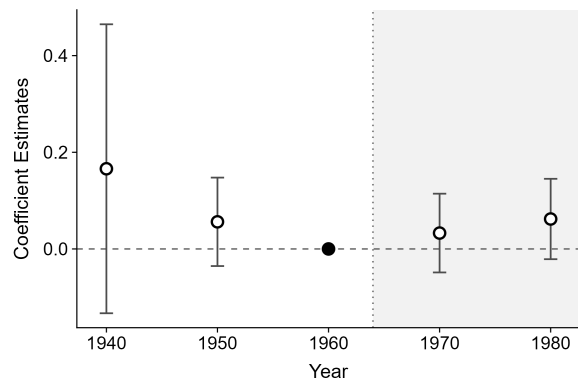
(c) Highest Grade Attended

Figure A.7: Event Study: Schooling Outcomes, Boys

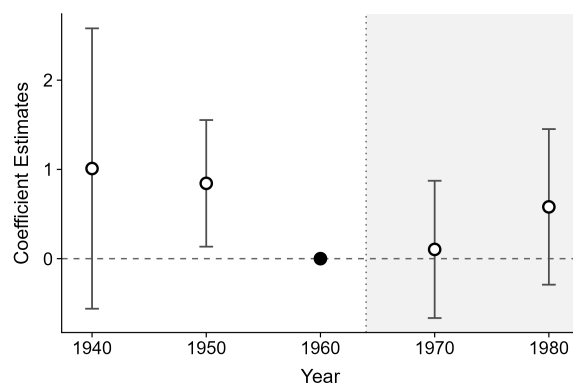
Notes: Event study estimates from equation (2) for boys aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. Post-termination coefficients for 1970 and 1980 capture the dynamic response to Bracero termination. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.



(a) School Attendance



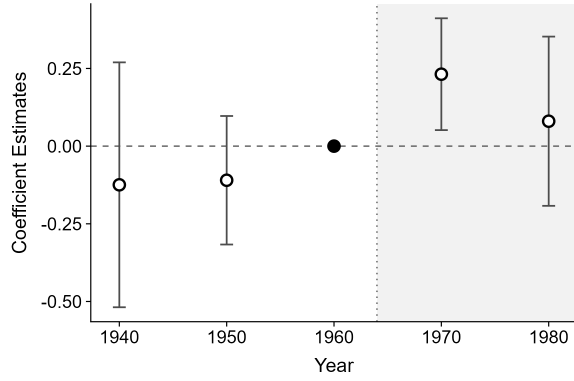
(b) Grade for Age



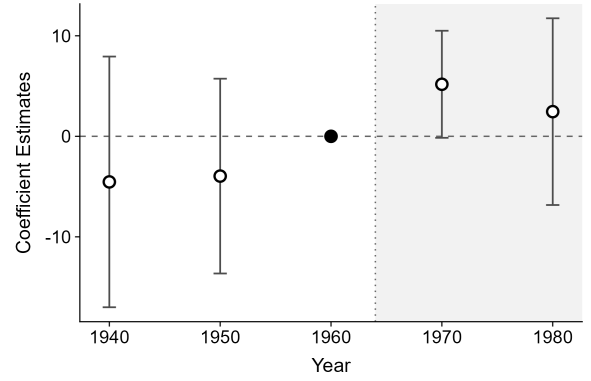
(c) Highest Grade Attended

Figure A.8: Event Study: Schooling Outcomes, Girls

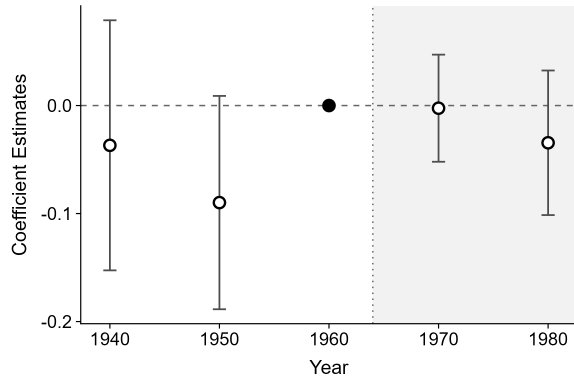
Notes: Event study estimates from equation (2) for girls aged 14-17 in farm households. The treatment variable is the 1955 state-level Bracero share $\bar{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. All post-termination coefficients are statistically indistinguishable from zero, consistent with no effect of Bracero termination on schooling outcomes. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.



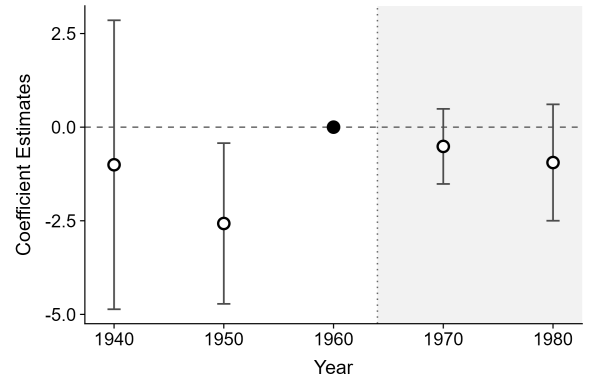
(a) Any Work Hours on Farm Binary (Boys)



(b) Hours Worked on Farm (Boys)



(c) Any Work Hours on Farm Binary (Girls)



(d) Hours Worked on Farm (Girls)

Figure A.9: Event Study: Hours Worked on Farm by Gender

Notes: Event study estimates from equation (2) for children aged 14-17 in farm households, estimated separately by gender. The treatment variable is the 1955 state-level Bracero share $\overline{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, supporting the parallel trends assumption. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.

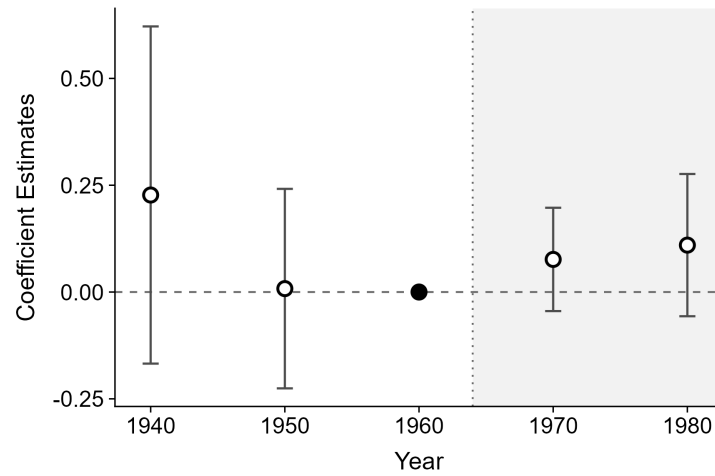


Figure A.10: Event Study: Farm Household Status

Notes: Event study estimates for the probability of residing in a farm household among U.S.-born children aged 14-17. The treatment variable is the 1955 state-level Bracero share $\overline{m}_s$ interacted with Census year indicators. The reference year is 1960, the last pre-termination Census year. Pre-termination coefficients for 1940 and 1950 are statistically indistinguishable from zero, and post-termination coefficients for 1970 and 1980 are also statistically indistinguishable from zero, providing no evidence of differential selection into or out of farm households following Bracero termination. Standard errors clustered at the state level. Vertical bars represent 95 percent confidence intervals.

Table A.1: Foreign Share of Seasonal Work by Crop (1964)

Crop	Foreign Share of Seasonal Workers (%)
Apples	3.8
Asparagus	19.0
Beans	2.4
Celery	32.4
Citrus	21.6
Cotton	3.7
Cucumbers	27.4
Grapes	3.3
Lettuce	55.3
Melons	28.4
Potatoes	3.6
Strawberries	13.8
Sugar beets	19.9
Sugarcane	46.9
Tobacco	1.9
Tomatoes	26.2
Median	19.4

Notes: Foreign share of seasonal workers by crop in 1964. Values are percentages. Data are from San (2023), Table 1, which is constructed using crop-level tabulations from the U.S. Department of Labor Bureau of Employment Security (1966), *The Mexican Farm Labor Program: 1942-1964*, Table 5.

Table A.2: Pre-Termination Bracero Exposure by State

State	Bracero Share 1955	State	Bracero Share 1955
New Mexico	0.609	Montana	0.025
Nebraska	0.323	Indiana	0.017
Arizona	0.269	Missouri	0.015
Texas	0.241	Idaho	0.012
California	0.231	Minnesota	0.010
South Dakota	0.212	Wisconsin	0.008
Nevada	0.090	Illinois	0.007
Arkansas	0.080	Tennessee	0.006
Wyoming	0.050	Washington	0.006
Colorado	0.045	Oregon	0.005
Michigan	0.034	Georgia	0.003
Utah	0.026	All other states	0.000

Notes: State-level Bracero exposure measured as the fraction of hired seasonal farm workers who were Mexican (Bracero) workers in 1955. Data are from archival administrative records compiled by Clemens et al. (2018). States not listed had no recorded Bracero workers in 1955 administrative records.

Table A.3: Effect of Bracero Termination by Age

	(1) Age 14	(2) Age 15	(3) Age 16	(4) Age 17
<i>Panel A: Farm Laborer Indicator</i>				
Treatment	0.102** (0.050)	0.132* (0.078)	0.181* (0.095)	0.166* (0.083)
Observations	30,304	30,394	30,729	29,499
<i>Panel B: Any Work Hours (Binary)</i>				
Treatment	0.096 (0.058)	0.130* (0.073)	0.257** (0.100)	0.260** (0.117)
Observations	30,304	30,394	30,729	29,499
<i>Panel C: Hours Worked (Continuous)</i>				
Treatment	3.287* (1.855)	4.769 (2.914)	5.679 (3.932)	8.998* (4.423)
Observations	30,304	30,394	30,729	29,499
<i>Panel D: School Attendance</i>				
Treatment	-0.118 (0.082)	-0.104 (0.103)	-0.094 (0.092)	-0.209** (0.073)
Observations	19,979	20,530	21,154	20,539
<i>Panel E: Highest Grade Attended</i>				
Treatment	-0.318 (0.575)	-0.530 (0.597)	-0.720 (0.694)	-0.494 (0.968)
Observations	30,304	30,394	30,729	29,499
<i>Panel F: Grade for Age</i>				
Treatment	-0.034 (0.086)	-0.024 (0.079)	-0.103 (0.106)	-0.072 (0.160)
Observations	30,304	30,394	30,729	29,499
Individual Controls	Yes	Yes	Yes	Yes
Household Head Controls	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children in farm households across five Census years (1940, 1950, 1960, 1970, 1980). The treatment variable is the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). Labor supply outcomes use person weights scaled by 0.5 for 1970 observations; schooling outcomes restrict the 1970 sample to Form 2 only and use original Census person weights. Controls include family size, female indicator, race/ethnicity indicators, birth order, and household head age, sex, marital status, and highest grade attended. Age and age squared are excluded since there is no within-age variation. Each column restricts the sample to a single age group: column (1) age 14, column (2) age 15, column (3) age 16, and column (4) age 17. Panels A-C report labor supply outcomes: farm laborer indicator, any positive work hours indicator, and hours worked last week in levels. Panels D-F report schooling outcomes: school attendance indicator, highest grade attended, and grade-for-age index. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.4: Effect of Bracero Termination on Farm Household Status

	(1) No Controls	(2) With Controls	(3) 1960-1970 Only
<i>Panel A: Farm Household Indicator</i>			
Treatment	0.031 (0.165)	0.027 (0.158)	0.081 (0.064)
Individual Controls	No	Yes	Yes
Household Head Controls	No	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	1,398,243	1,398,243	408,435

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample includes all U.S.-born children aged 14-17 across five Census years (1940, 1950, 1960, 1970, 1980), not restricted to farm households. Column (3) restricts the sample to the 1960 and 1970 Census years only. The treatment variable is the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). Person weights are scaled by 0.5 for 1970 observations to reflect the pooling of Form 1 and Form 2 state samples. Controls include family size, age, age squared, female indicator, race/ethnicity indicators, and birth order. Head controls include household head age, sex, marital status, and highest grade attended. The outcome is an indicator equal to one if the child resides in a farm household. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.5: Robustness: Alternative Specifications

	(1) Baseline	(2) Age Fixed Effects	(3) No Head Controls	(4) Drop CA & TX
<i>Panel A: Farm Laborer Indicator</i>				
Treatment	0.146** (0.069)	0.146** (0.069)	0.141** (0.067)	0.120* (0.060)
Observations	120,926	120,926	120,926	111,970
<i>Panel B: Any Work Hours (Binary)</i>				
Treatment	0.189** (0.077)	0.189** (0.077)	0.186** (0.075)	0.121** (0.051)
Observations	120,926	120,926	120,926	111,970
<i>Panel C: Hours Worked (Continuous)</i>				
Treatment	5.700* (3.022)	5.701* (3.022)	5.536* (2.926)	3.003 (1.827)
Observations	120,926	120,926	120,926	111,970
<i>Panel D: School Attendance</i>				
Treatment	-0.134* (0.079)	-0.134* (0.079)	-0.122* (0.073)	-0.066 (0.057)
Observations	82,202	82,202	82,202	75,921
<i>Panel E: Highest Grade Attended</i>				
Treatment	-0.525 (0.656)	-0.525 (0.656)	-0.528 (0.658)	-0.720 (0.840)
Observations	120,926	120,926	120,926	111,970
<i>Panel F: Grade for Age</i>				
Treatment	-0.059 (0.102)	-0.059 (0.102)	-0.056 (0.102)	-0.112 (0.096)
Observations	120,926	120,926	120,926	111,970
Individual Controls	Yes	Yes	Yes	Yes
Household Head Controls	Yes	Yes	No	Yes
State Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Age Fixed Effects	No	Yes	No	No

Notes: Continuous difference-in-differences estimates based on Equation 1 with state and year fixed effects. Sample restricted to U.S.-born children aged 14-17 in farm households across five Census years (1940, 1950, 1960, 1970, 1980). The treatment variable is the interaction of the 1955 state-level Bracero share ($\bar{m}_s$) with an indicator for post-1964 Census years (1970 and 1980). Labor supply outcomes use person weights scaled by 0.5 for 1970 observations; schooling outcomes restrict the 1970 sample to Form 2 only and use original Census person weights. Controls include family size, age, age squared, female indicator, race/ethnicity indicators, birth order, and household head age, sex, marital status, and highest grade attended. Column (1) reproduces the baseline specification. Column (2) replaces the age polynomial with age fixed effects. Column (3) excludes household head controls. Column (4) drops California and Texas, the two states with the largest absolute Bracero worker counts, to verify results are not driven by these dominant states. Panels A-C report labor supply outcomes: farm laborer indicator, any positive work hours indicator, and hours worked last week in levels. Panels D-F report schooling outcomes: school attendance indicator, highest grade attended, and grade-for-age index. Standard errors clustered at the state level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.